

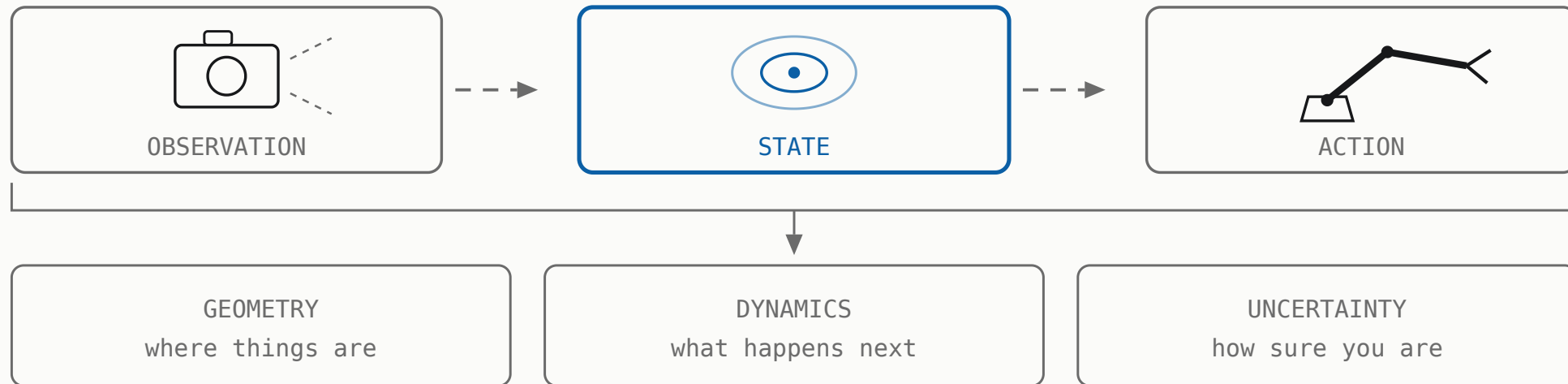
Introduction and Linear Algebra I

Spaces, Maps, Rank, Solutions

Mathematics for Robotics · ROB-GY 6013 · NYU Tandon

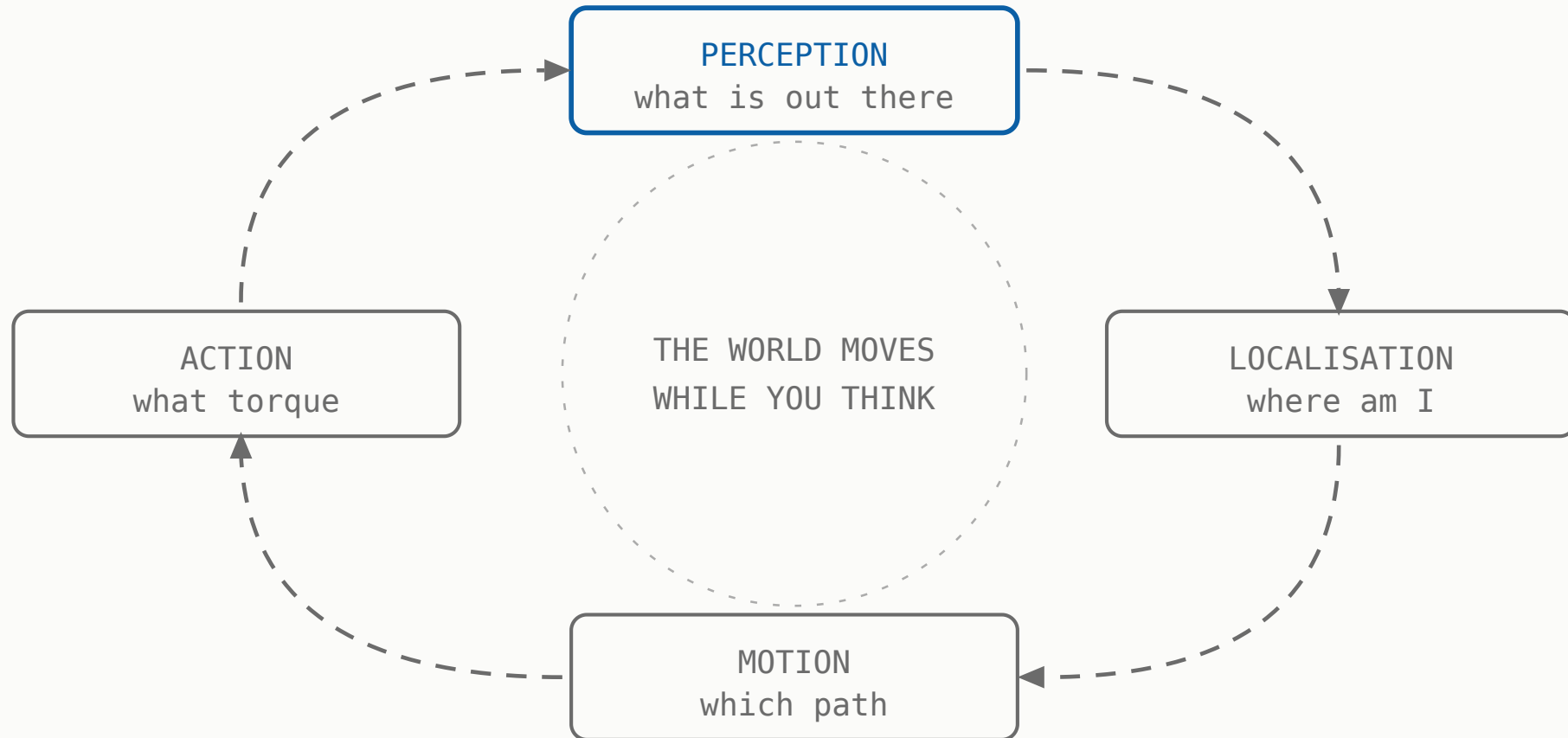
Jing Zhang · Fall 2026 · September 2

Why Mathematics for Robotics?



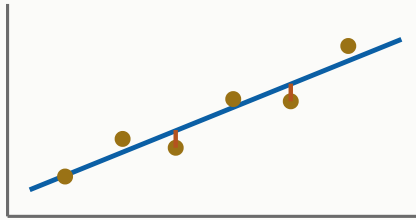
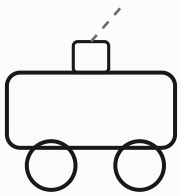
The three panels are recurring questions, not one per box above.

Why Mathematics for Robotics?



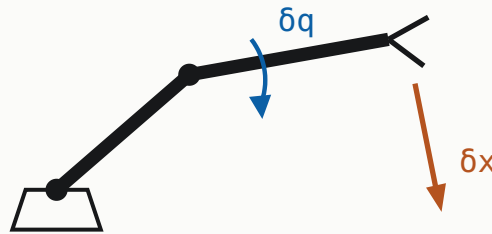
The loop closes many times a second.

Why Mathematics for Robotics?



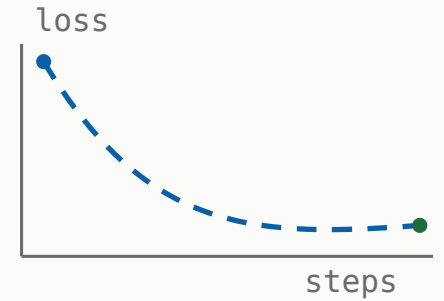
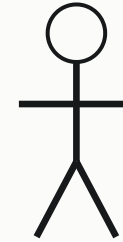
CALIBRATION

locally, $Ax \approx b$



ROBOT MOTION

$\delta x \approx J \delta q$



LEARNING

$\arg \min_{\theta} \mathcal{L}(\theta)$

Different surface, same mathematical structure.

Why Mathematics for Robotics?

Robotics changes quickly.

The mathematical structure changes much more slowly.

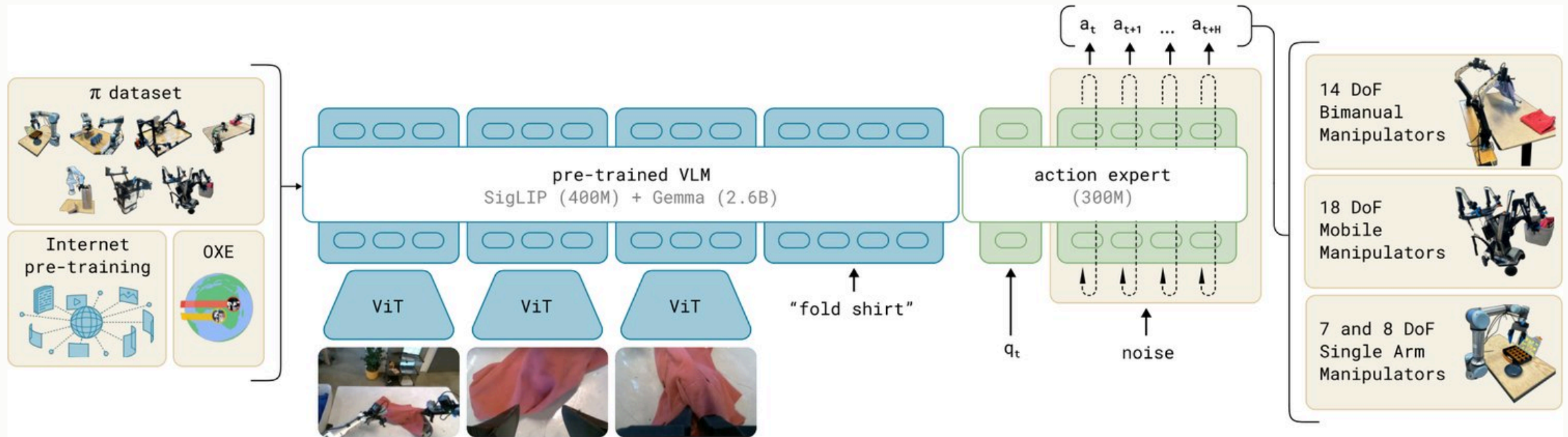
Why Mathematics in the GenAI Era?



Google DeepMind · Advanced dexterity with Gemini Robotics 2, July 2026

[opens on YouTube](#)

Why Mathematics in the GenAI Era?



VLM · attention, masks · **Transformers**

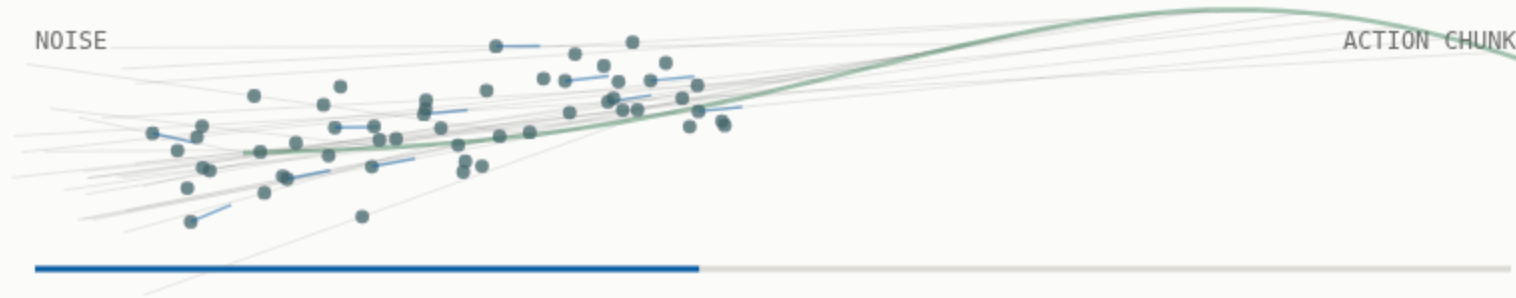
q_t · vector spaces · **Linear Algebra I-IV**

noise $\rightarrow a_t$ · a vector field · **ODE** · **Flow Matching**

A modern VLA is not replacing mathematics. It is built from mathematics.

Black, K. et al. *no: A Vision-Language-Action Flow Model for General Robot Control*, Figure 3. RSS, 2025. arXiv:2410.24164.

Why Mathematics in the GenAI Era?



$$\boxed{\frac{dA_\tau}{d\tau} = v_\theta(A_\tau, o, \tau)} \quad A_0 = \epsilon, \quad \hat{a} = A_1$$

τ flow time, 0 to 1; the field changes with it

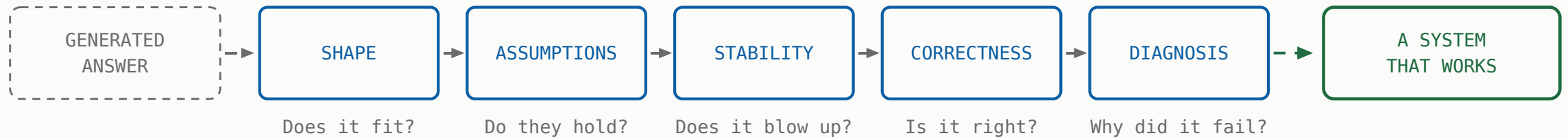
v_θ the trained vector field

A_0 the start: pure noise ϵ

$\hat{a} = A_1$ the action chunk, read off at $\tau = 1$

Black, K. et al. *Ho*. RSS, 2025. arXiv:2410.24164.

Why Mathematics in the GenAI Era?



AI can generate. You still have to judge.

Why Mathematics in the GenAI Era?

WHAT AN ML INTERVIEW ASKS ABOUT

Rank / span ●	Null / image ●	Independence ●
Inverse ●	Singular ●	Orthogonality
Positive semidefinite	Eigenvalues	Jacobian
Hessian	Determinant	Dot product

AND TO WRITE, BY HAND

Lectures 5 · 6 · 10

transformer end-to-end	causal, cross, self attention	flash attention
attention backward pass	MLP forward and backward	SGD training loop

“...with AI assistance completely off.”

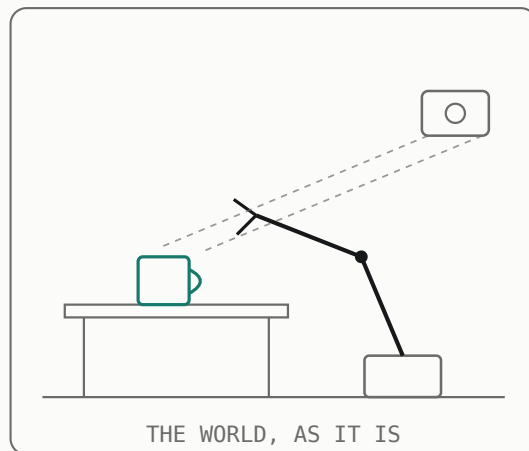
Sapora, S. *ML Job Interviews: The Ultimate Guide*, 2026 · silviasapora.github.io/blog/ml-interviews.html
Liu, A. *Notes on the Industry Job Search*, 2026 · alisawuffles.github.io/blog/job-search/

The Math We Keep Coming Back To

x	state	— — —▶	Linear Algebra I · ODE · Probability I
A	linear map	— — —▶	Linear Algebra I–IV
f	nonlinear function	— — —▶	ML Math I · ODE
J	Jacobian	— — —▶	Linear Algebra IV · ML Math I–II
p	probability distribution	— — —▶	Probability I–II · Diffusion
L	loss	— — —▶	ML Math II · Diffusion · Flow Matching

Different problems, the same mathematical ideas.

The Math We Keep Coming Back To



REPRESENT
----->

STATE VECTOR

$$\begin{bmatrix} 0.42 \\ -1.05 \\ 0.90 \\ 0.13 \\ 2.71 \end{bmatrix}$$

JUST NUMBERS

FOUR ROBOT QUANTITIES

$$q \in \mathbb{R}^7$$

joint configuration

$$p \in \mathbb{R}^3$$

position

$$I \in \mathbb{R}^{3 \times H \times W}$$

image

$$a \in \mathbb{R}^{16 \times 7}$$

action chunk

Each is a vector, or an array that can be vectorized.

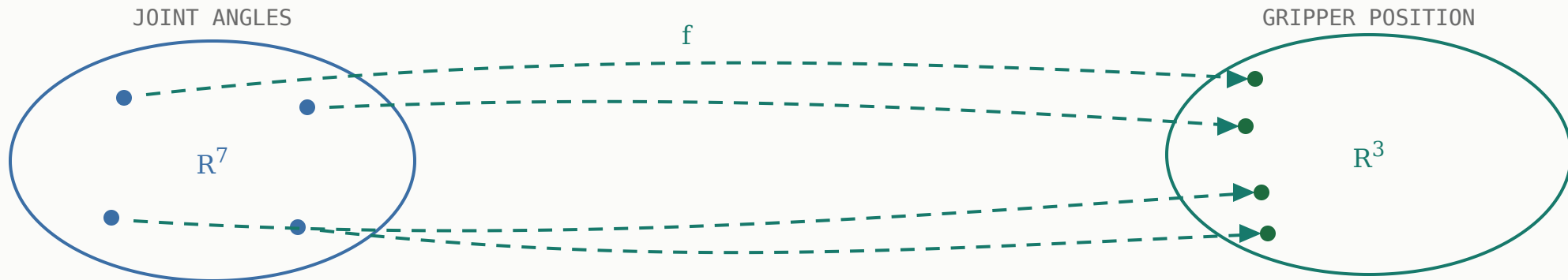
The Math We Keep Coming Back To

MAP $f : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad u \mapsto y = f(u)$

domain $\mathbb{R}^n$, the allowed inputs

codomain $\mathbb{R}^m$, the space the outputs live in

$u \mapsto y$ the rule that carries one to the other



ONE RULE, AND NO INPUT GETS TWO ANSWERS

Forward kinematics: $\mathbb{R}^7 \rightarrow \mathbb{R}^3$, joint angles to gripper position.

The Math We Keep Coming Back To

LOCAL LINEARISATION

$$f(x + \delta x) \approx f(x) + J(x) \delta x$$

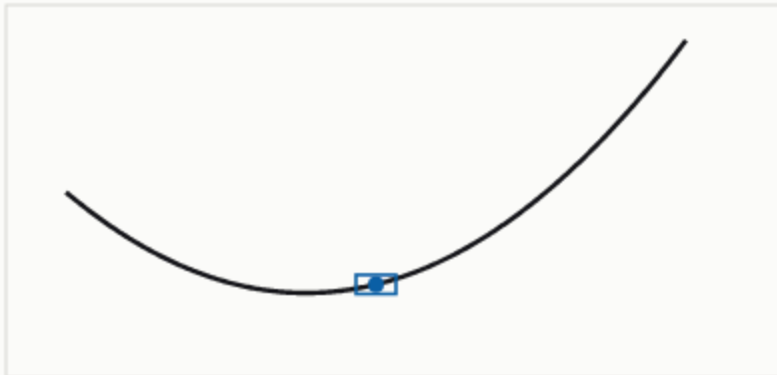
x the current point

δx a small change

$J(x)$ the Jacobian, $J(x) \in \mathbb{R}^{m \times n}$

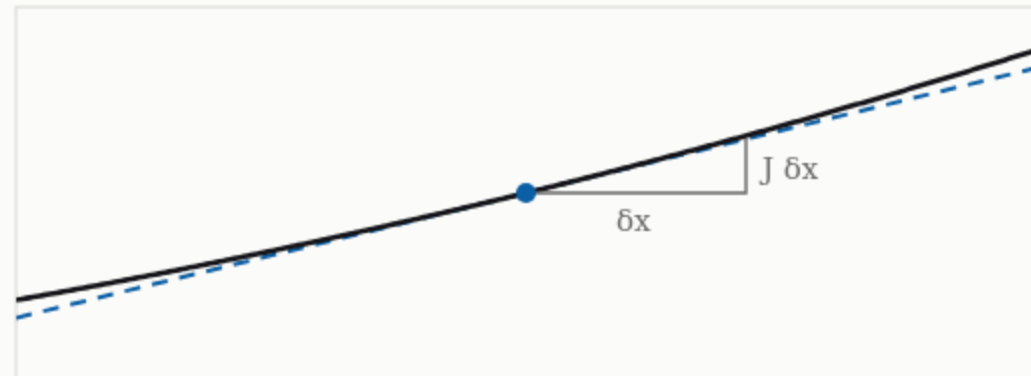
Many nonlinear robotics problems become linear when viewed locally.

f , the whole map



the tangent: this is J

zoom $\times 7$



Left, the shrinking window. Right, what is inside it.

The Math We Keep Coming Back To

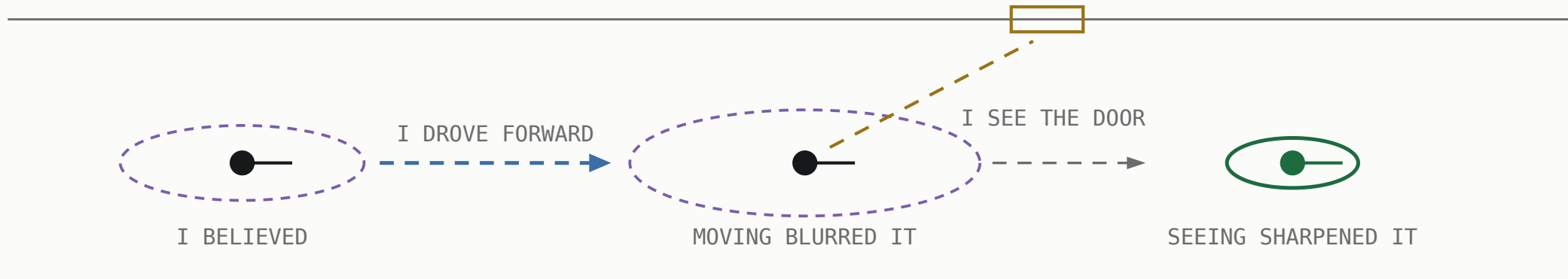
WHERE AM I? $x_t = f(x_{t-1}, u_t) + w_t, \quad z_t = h(x_t) + v_t$

x_t the latent state, usually not observed directly

u_t, z_t the command, the measurement

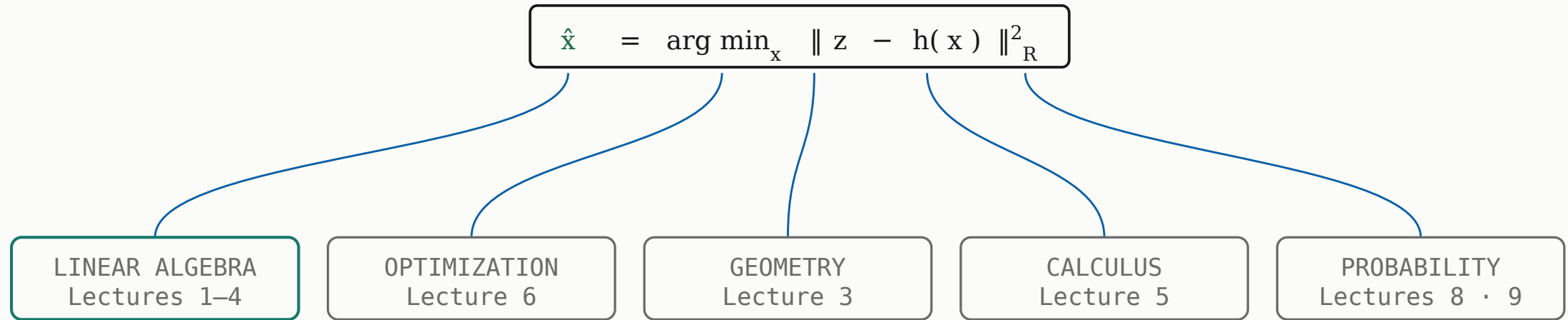
f, h motion and sensor models

w_t, v_t noise



Thrun, S., Burgard, W., Fox, D. *Probabilistic Robotics*, Ch. 7. MIT Press, 2005.

The Math We Keep Coming Back To



$\|e\|_R^2 = e^T R^{-1} e$, with R the measurement covariance.

- e $z - h(x)$ here, measurement minus prediction, which the course writes r ; from Lecture 5 on, e is prediction minus target, so the two differ by a sign

Barfoot, T. D. *State Estimation for Robotics*, Chapter 4. Cambridge University Press, 2017.

Course information

THE MEETING

WHEN	Wednesdays 5:00–7:30 PM
WHERE	2 MetroTech Center, Room 909
ONLINE	nyu.zoom.us/j/95222620311
UPDATES	slides and recordings on the course page
QUESTIONS	Ed Discussion, linked from the course page

THE STAFF

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Shichen Zhang

teaching assistant · sz4968@nyu.edu

office hours to be announced

Everything on these four pages, plus the schedule and the reading: jingz6676.github.io/rob6013.html

Course information

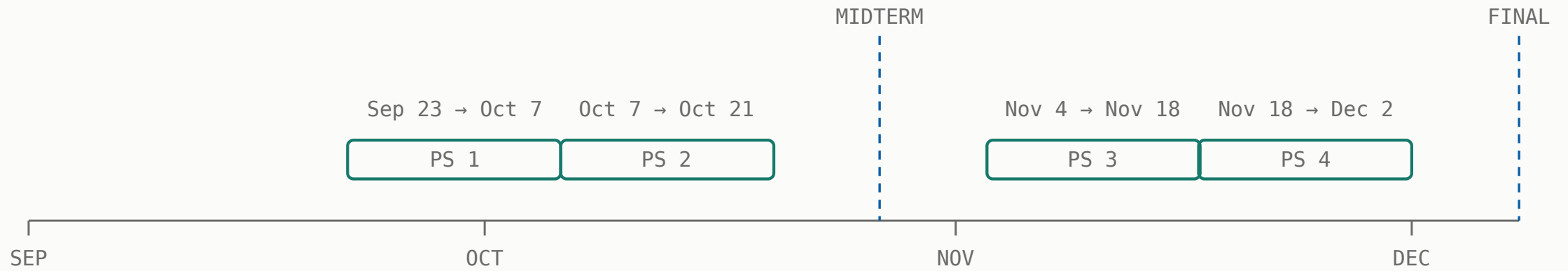
GRADING

40%	25%	35%
FOUR PROBLEM SETS understanding, not recall	MIDTERM 28 October, in person	FINAL 9 December, in person

up to +10% on the course grade, for sets typeset in LaTeX or Markdown

Course information

PROBLEM SETS AND EXAMINATIONS · RELEASED → DUE



One sheet of paper, both sides, prepared by you, and a scientific calculator.
No phone, laptop, tablet, or watch.

Course information

POLICIES

LATE WORK	10% of the score per calendar day; closed once solutions are posted
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COLLABORATION	discuss with anyone, then write your solution alone
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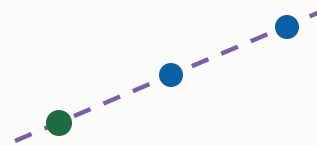
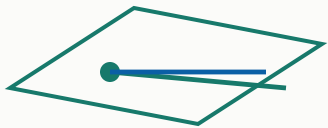
USING AI	any tool to understand it, none to write your solutions
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ACCOMMODATIONS	through the Moses Center, with no need to tell me why
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IF IT GOES WRONG	email me and Shichen
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LINEAR ALGEBRA I

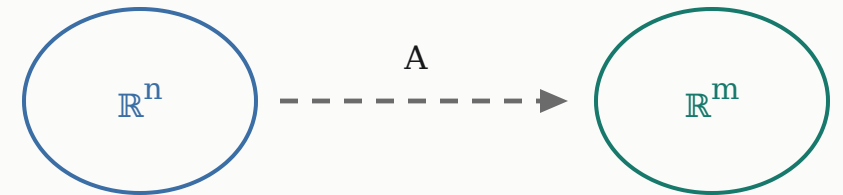
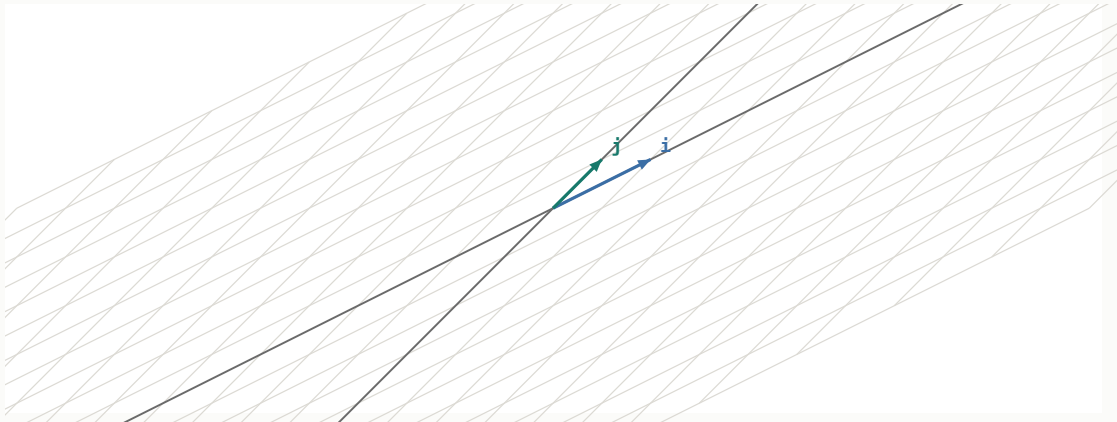
Spaces, Maps, Rank, Solutions



A matrix is a map

$$A \in \mathbb{R}^{m \times n}, \quad x \in \mathbb{R}^n \mapsto Ax \in \mathbb{R}^m$$

- A the map itself, m rows by n columns
- x the input, a column of n numbers
- Ax the output, a column of m numbers
- n input dimension: the number of columns
- m output dimension: the number of rows; m and n need not be equal



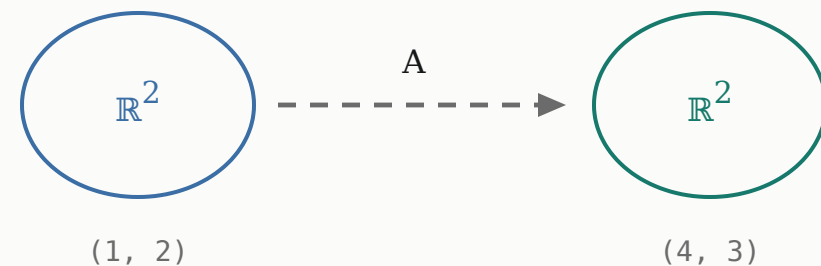
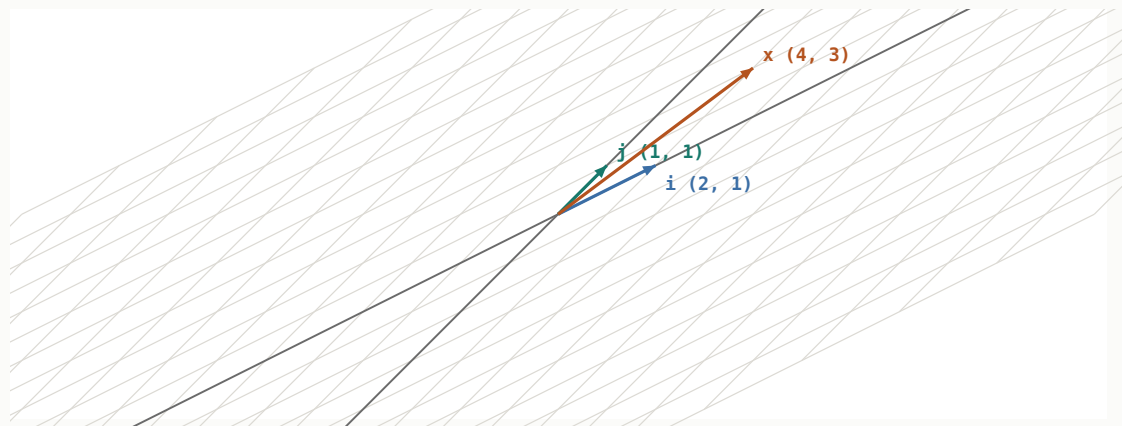
Sanderson, G. *Essence of Linear Algebra*, ch. 3. 3Blue1Brown, 2016.

A matrix is a map

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+2 \\ 1+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ the first column, where e_1 lands

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ the second column, where e_2 lands



A matrix is a map

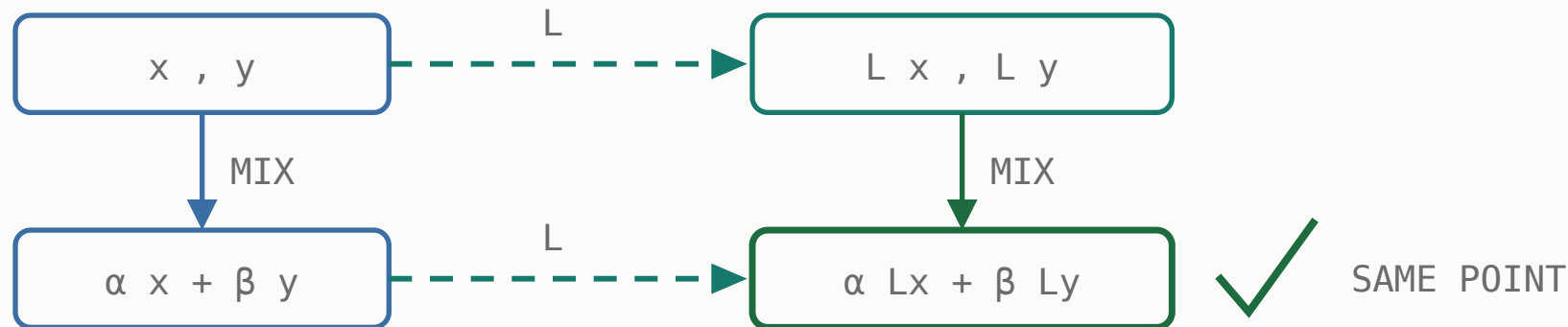
LINEAR MAP

$$L(\alpha x + \beta y) = \alpha L(x) + \beta L(y)$$

additivity $L(x + y) = L(x) + L(y)$, the case $\alpha = \beta = 1$

homogeneity $L(\alpha x) = \alpha L(x)$, the case $\beta = 0$

superposition both at once, for all x, y and all $\alpha, \beta \in \mathbb{R}$

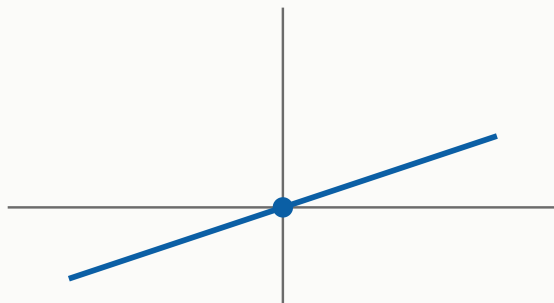


Axler, S. *Linear Algebra Done Right*, 4th ed., Definition 3.1, p. 52. Springer, 2024. Open access.

QUESTIONS

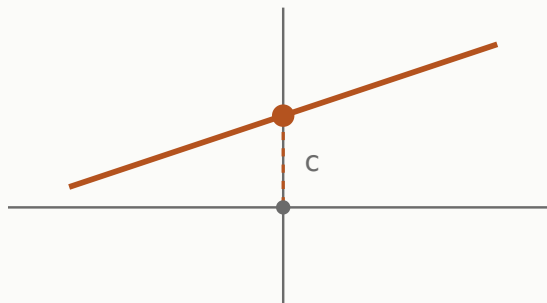
Which of these three maps is linear?

$$f(x) = a x$$



LINEAR

$$f(x) = a x + c, \quad c \neq 0$$

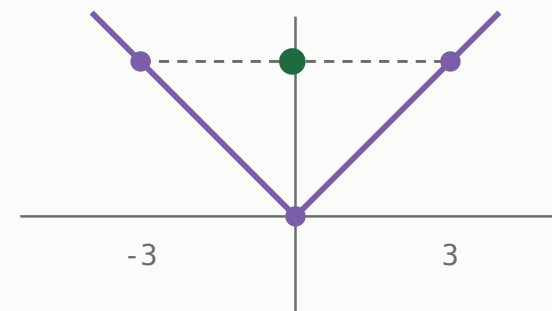


AFFINE, NOT LINEAR

$$a = 1, \quad c = 2$$

✗ $f(0) = 2 \neq 0$

$$f(x) = |x|$$



NOT LINEAR



$$f(0) = 0$$



$$f(-3) = 3, \quad -f(3) = -3$$

A matrix is a map

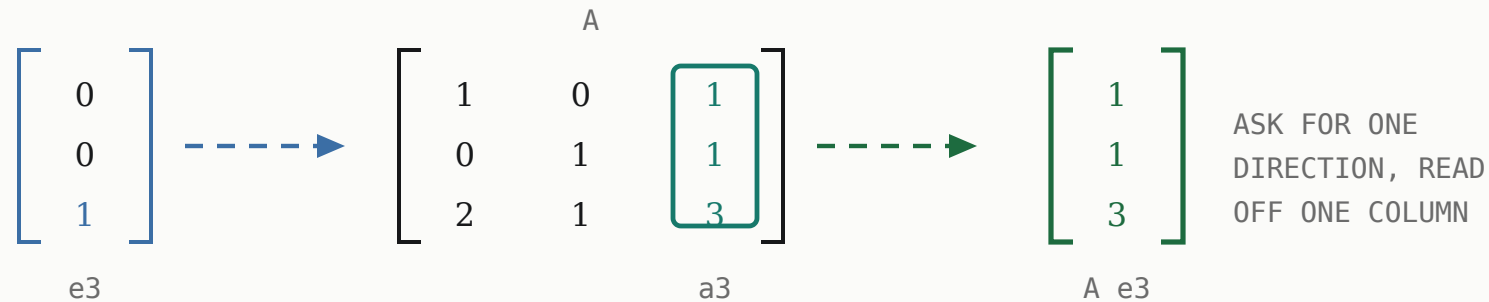
$$e_j = (0, \dots, 1, \dots, 0)^T$$

the 1 sits in slot j ; every other slot holds 0

COLUMN RULE

$$Ae_j = a_j$$

a_j column j of A , a whole output vector



Axler, S. *Linear Algebra Done Right*, 4th ed.: standard basis 2.27(a), p. 39; columns 3.50, p. 76. Springer, 2024.

A matrix is a map

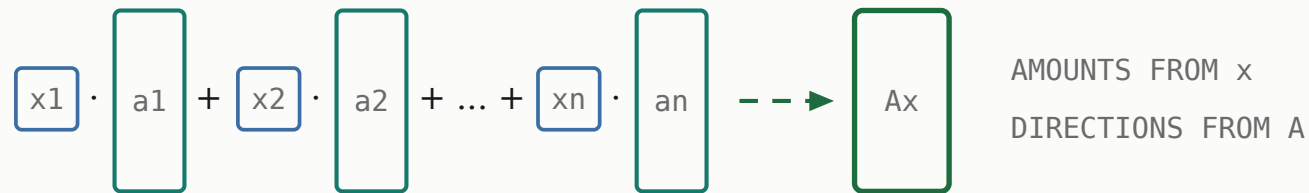
LINEAR COMBINATION OF COLUMNS

$$Ax = \sum_{j=1}^n x_j (Ae_j) = x_1 a_1 + \cdots + x_n a_n$$

x_j coordinate j of x , used as an amount

Ae_j column j , from the page before

a_j column j of A , a whole direction



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.50 linear combination of columns, p. 76. Springer, 2024.

A matrix is a map

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = 3a_1 + 2a_2 = \begin{bmatrix} 3 \\ 2 \\ 8 \end{bmatrix}$$

a_1, a_2 (1,0,2) and (0,1,1), the first two columns

3, 2, 0 the amounts asked of the three columns

$$3 \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}_{a_1} + 2 \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}_{a_2} + 0 \cdot \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}_{a_3} = \begin{bmatrix} 3 \\ 2 \\ 8 \end{bmatrix}_{Ax}$$

THREE COLUMNS
THREE AMOUNTS
ONE POINT

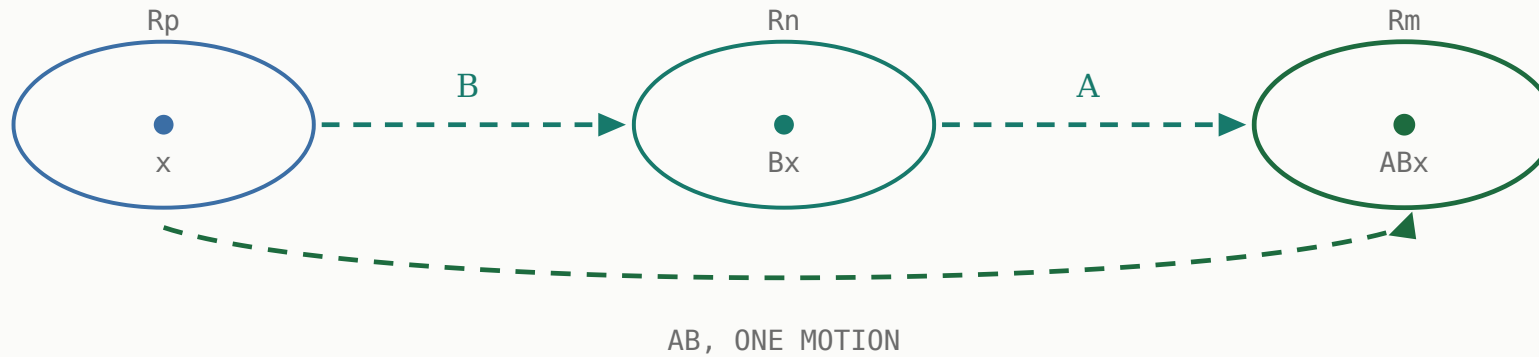
A matrix is a map

DO B FIRST, THEN A $(AB)x = A(Bx)$

shapes $B \in \mathbb{R}^{n \times p}$, $A \in \mathbb{R}^{m \times n}$

Bx where B sends x first

$A(Bx)$ where A then sends that



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.43 matrix of product of linear maps, p. 74. Springer, 2024.

A matrix is a map

TRANSPOSE $(A^T)_{k,j} = A_{j,k}$

row j of A becomes column j of A^T

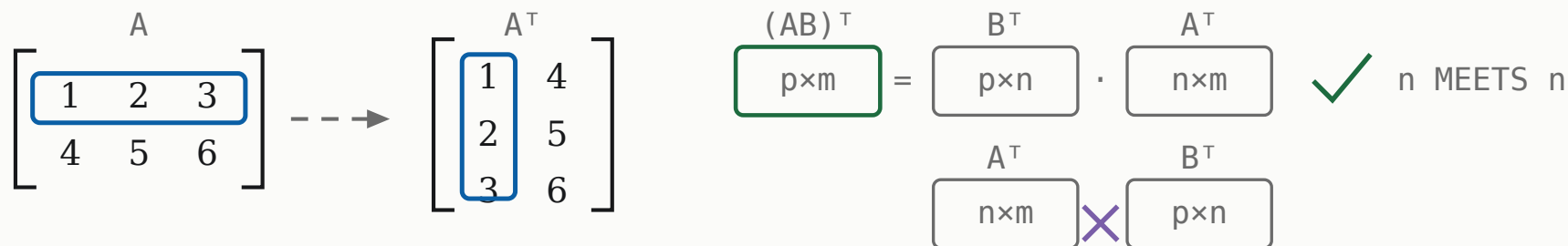
$m \times n$ becomes $n \times m$

IT REVERSES A PRODUCT

$$(AB)^T = B^T A^T$$

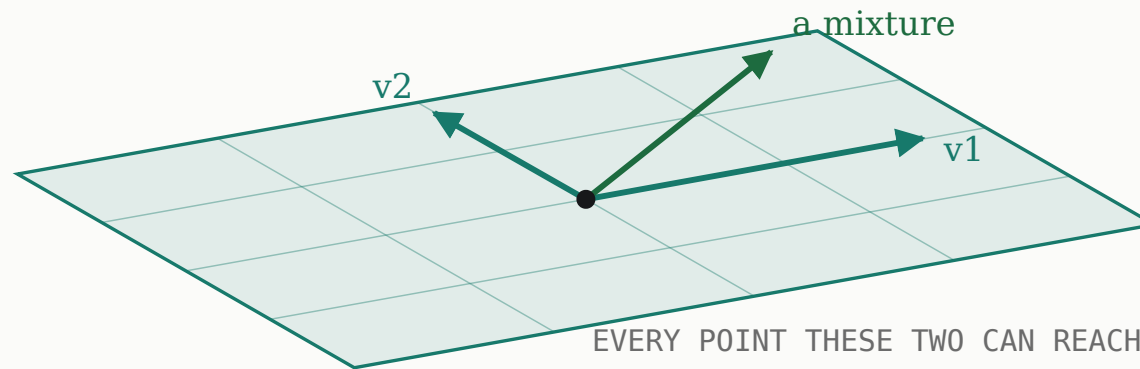
$(AB)^T$ $p \times m$, the shape the answer must have

$B^T A^T$ $p \times n$ times $n \times m$, so n meets n



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.54 transpose, p. 77; $(AC)^t = C^t A^t$, p. 78. Springer, 2024.

Span and column space



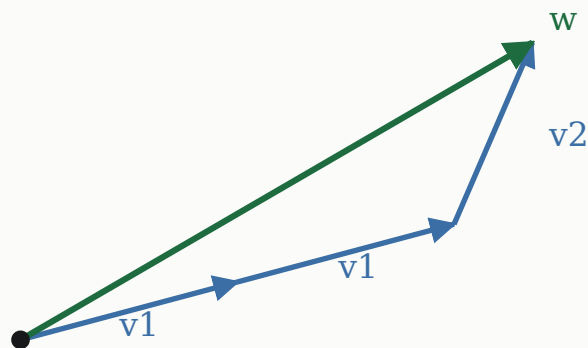
Span and column space

LINEAR COMBINATION $w = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k$

v_i the directions you are handed

c_i any reals, negative and zero included

w where those weights land you



TWO STEPS OF v_1

ONE STEP OF v_2

$$w = 2 v_1 + v_2$$

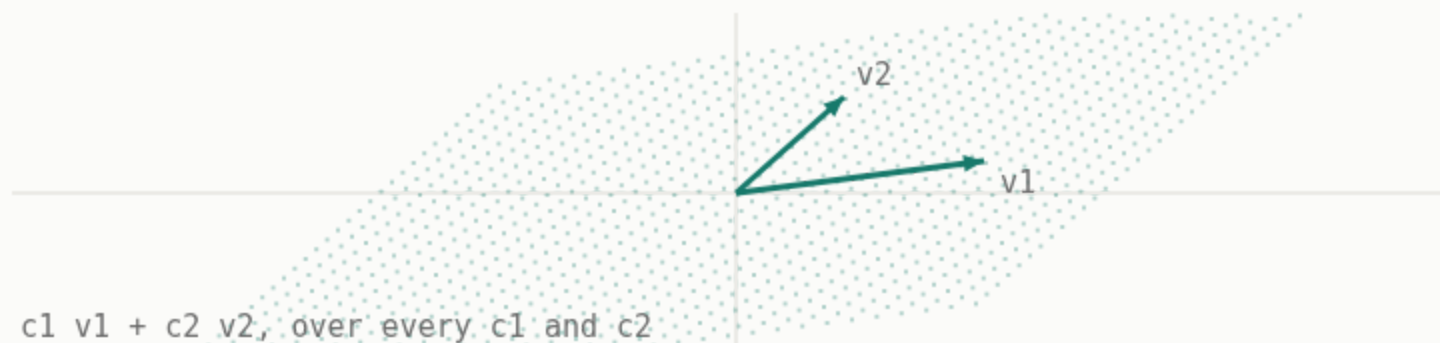
Axler, S. *Linear Algebra Done Right*, 4th ed., 2.2 linear combination, p. 28. Springer, 2024.

Span and column space

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

SPAN $\text{span}(v_1, \dots, v_k) = \{c_1 v_1 + \dots + c_k v_k : c_i \in \mathbb{R}\}$

{ } every w the weights can reach, all at once



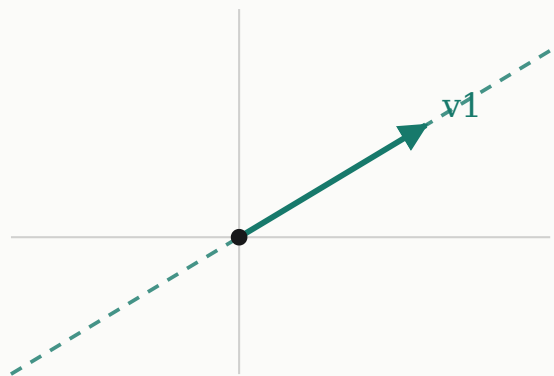
Let both weights roam: the plane fills in.

WORKED EXAMPLE

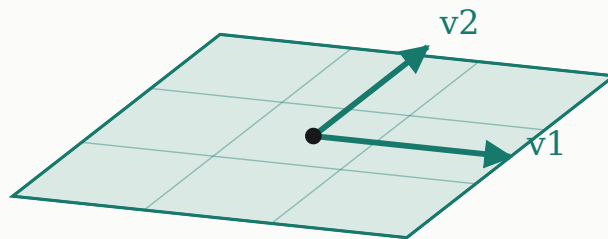
$v_1 = (1, 0, 2)$, $v_2 = (0, 1, 1)$: $3v_1 + 2v_2 = (3, 2, 8)$ is in the span.

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.4 span, p. 29. Springer, 2024.

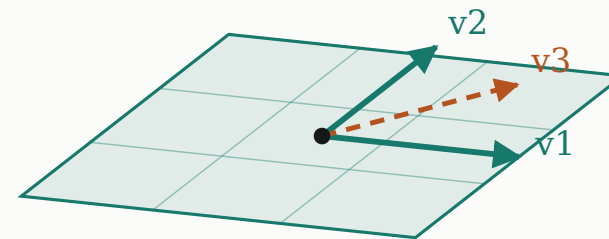
Span and column space



ONE NONZERO VECTOR · A LINE



TWO INDEPENDENT ONES · A PLANE



A THIRD IN IT · SAME PLANE

$$\text{span}(v_1) = \{c v_1\}$$

$$\text{span}(v_1, v_2) = \{c_1 v_1 + c_2 v_2\}$$

$$\text{span}(v_1, v_2, v_3) = \text{span}(v_1, v_2)$$

the conditions dependent pair: a line, or $\{0\}$ if both zero

v_3 already a mixture of the first two

Sanderson, G. *Essence of Linear Algebra*, Chapter 2. 3Blue1Brown, 2016.

Span and column space

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

$$b \in \text{span}(v_1, \dots, v_k) \iff Vc = b \text{ has a solution}$$

V the arrows as columns

c the unknown weights

b the target vector

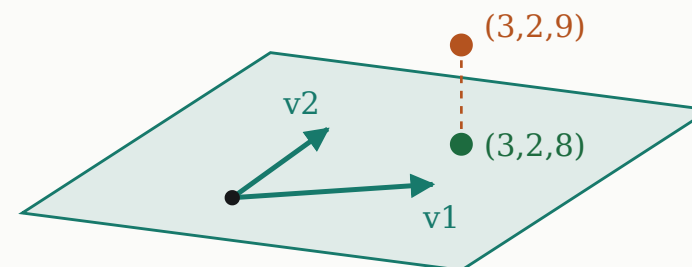
IS $(3, 2, 9)$ IN THE SPAN?

$$c_1(1, 0, 2) + c_2(0, 1, 1) = (3, 2, 9)$$

row 1 gives $c_1 = 3$, row 2 gives $c_2 = 2$

row 3 is then forced: $2(3) + 2 = 8$, not 9

span of v_1, v_2



One lands, one misses.

Span and column space

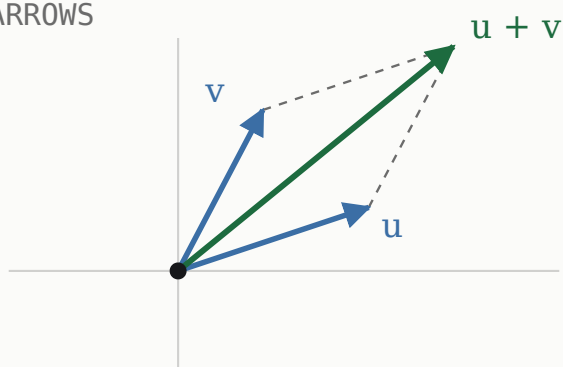
A vector space is a set equipped with vector addition and scalar multiplication satisfying the vector-space axioms.

WHAT WE USE

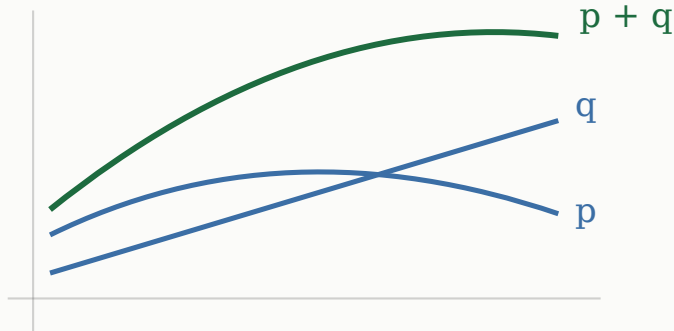
$$u, v \in V, \alpha, \beta \in \mathbb{R} \implies \alpha u + \beta v \in V$$

closure the one property this course leans on all term

ARROWS



POLYNOMIALS



MATRICES

$$\begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} + \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} = \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix}$$

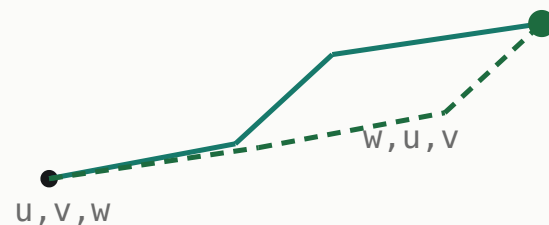
Axler, S. *Linear Algebra Done Right*, 4th ed., 1.20 vector space, p. 12. Springer, 2024.

Span and column space

ADDITION

$$\begin{aligned} u + v &= v + u & (u + v) + w &= u + (v + w) \\ v + 0 &= v & \exists w : v + w &= 0 \end{aligned}$$

0 and w an origin, and an undo for every vector



ANY ORDER, SAME POINT

SCALING

$$\begin{aligned} 1v &= v & \alpha(\beta v) &= (\alpha\beta)v \\ \alpha(u + v) &= \alpha u + \alpha v & (\alpha + \beta)v &= \alpha v + \beta v \end{aligned}$$

the payoff obey all eight and every theorem here applies

Axler, S. *Linear Algebra Done Right*, 4th ed., 1.20 vector space, p. 12. Springer, 2024.

Span and column space

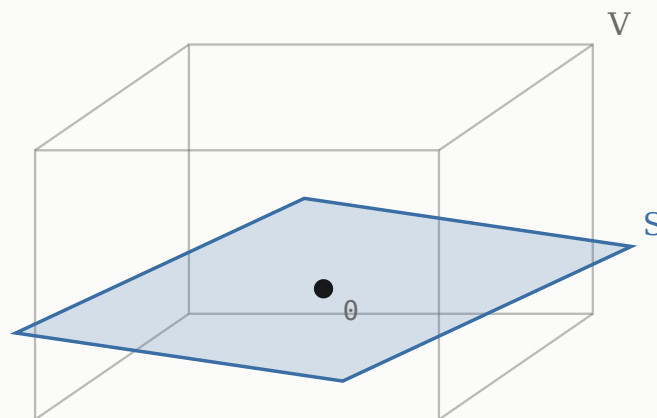
A subset $S \subseteq V$ is a subspace if it is itself a vector space under the same operations.

EQUIVALENT TEST

$$S \neq \emptyset \text{ and } u, v \in S, \alpha, \beta \in \mathbb{R} \Rightarrow \alpha u + \beta v \in S$$

[nonempty](#) plus closure already gives $0 \in S$: choose $\alpha = \beta = 0$

[next page](#) the same test, split into three checks



Axler, S. *Linear Algebra Done Right*, 4th ed., 1.33 subspace and 1.34 conditions for a subspace, p. 18. Springer, 2024.

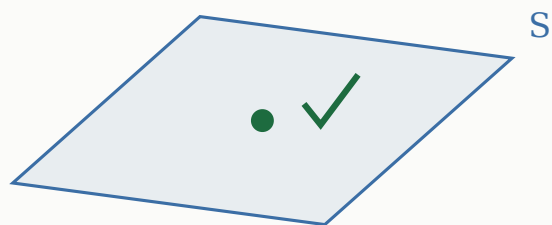
Span and column space

CONDITIONS FOR A SUBSPACE $0 \in S$; $u + v \in S$; $\alpha u \in S$

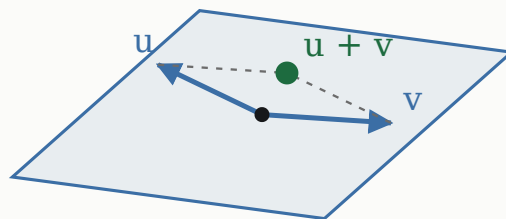
0 the zero vector is a member

$u+v$ any two members add to a member

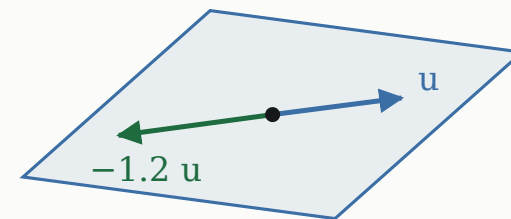
αu any multiple stays in, negatives included



0 IS IN S



$u + v$ IS IN S



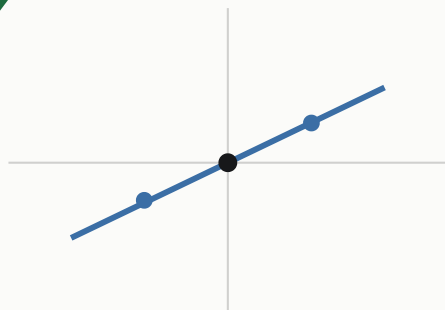
ANY MULTIPLE IS IN S

Axler, S. *Linear Algebra Done Right*, 4th ed., 1.34 conditions for a subspace, p. 18. Springer, 2024.

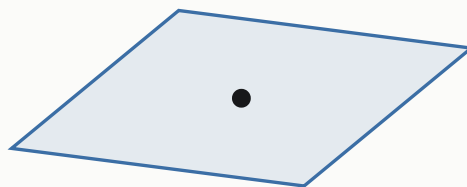
Span and column space

FASTEST DISQUALIFIER $0 \notin S \implies S$ is not a subspace

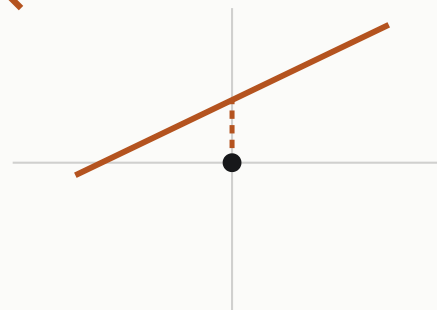
$0 \in S$ necessary, never sufficient: closure still has to hold



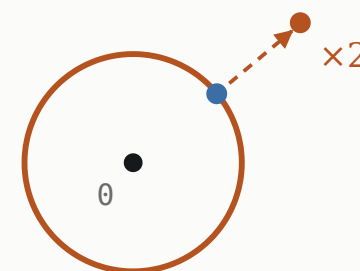
LINE THROUGH 0



PLANE THROUGH 0



SHIFTED LINE · AFFINE



CIRCLE

Axler, S. *Linear Algebra Done Right*, 4th ed., 1.35 example: subspaces, p. 19. Springer, 2024.

Span and column space

CLAIM

$v_1, \dots, v_k \in V \implies \text{span}(v_1, \dots, v_k)$ is a subspace of V

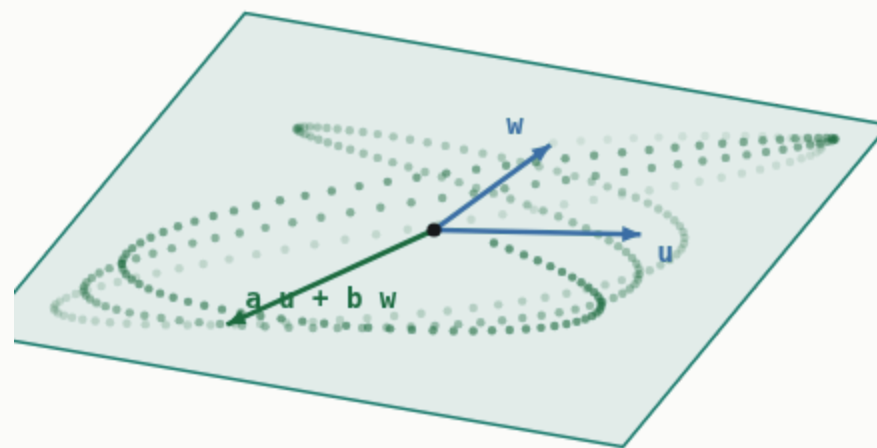
S every mixture of the list; the smallest subspace containing it

$$0 = \sum_{i=1}^k 0 v_i \in S$$

every $c_i = 0$ the origin is a member

$$\alpha u + \beta w = \sum_{i=1}^k (\alpha c_i + \beta d_i) v_i \in S$$

u, w any two members: $u = \sum_i c_i v_i, w = \sum_i d_i v_i$



Axler, S. **Linear Algebra Done Right**, 4th ed., 2.6 span is the smallest containing subspace, p. 29. Springer, 2024.

Span and column space

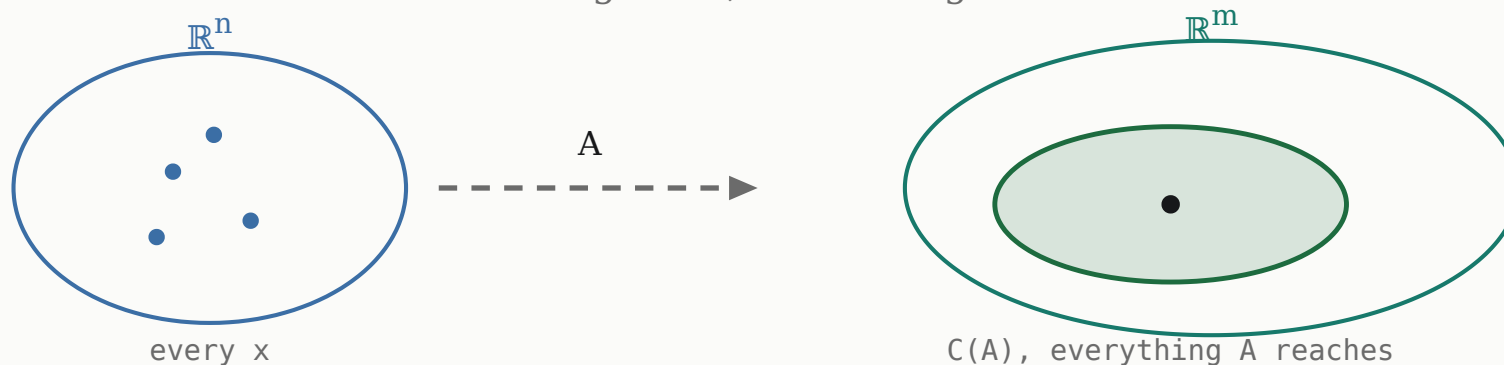
COLUMN SPACE

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$$

A any $m \times n$ matrix; x runs over the whole of $\mathbb{R}^n$

$\subseteq \mathbb{R}^m$ a subspace of the codomain; if $m = n$ the roles still differ

also called the image of A , or the range of A



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.16 range and 3.18 the range is a subspace, p. 61. Springer, 2024.

Span and column space

DEFINITION

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\}$$

x runs over every input

COLUMN VIEW

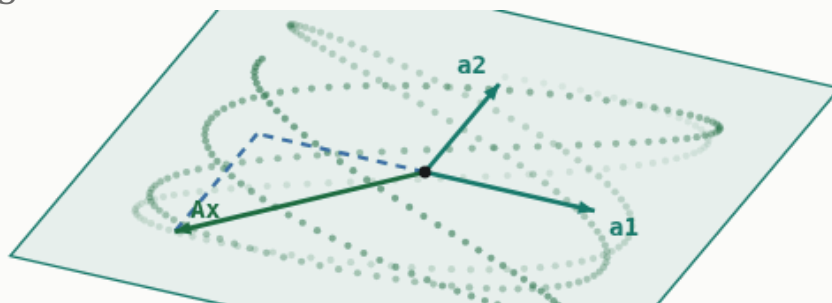
$$Ax = \sum_{j=1}^n x_j a_j$$

a_j column j of A



$$\mathcal{C}(A) = \text{span}(a_1, \dots, a_n)$$

one set reachable outputs and mixtures of the columns are the same thing



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.16 range, p. 61; 3.50 linear combination of columns, p. 76. Springer, 2024.

Span and column space

QUESTION $Ax = b$?

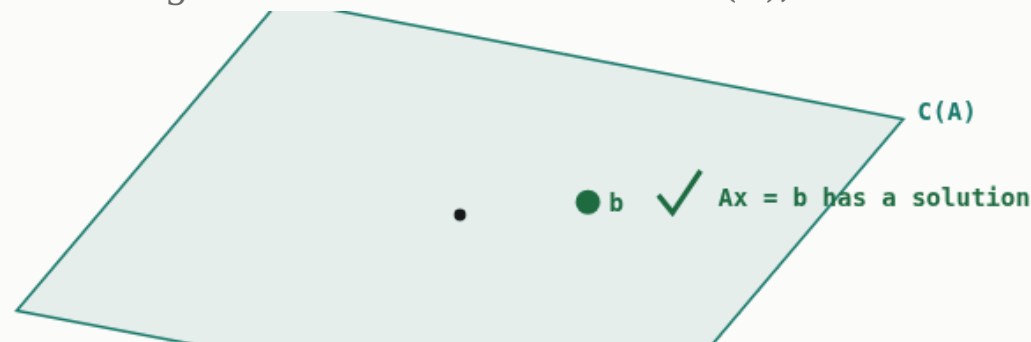
b a target on the output side

$$Ax = b \text{ has a solution} \iff \exists x : Ax = b \iff b \in \{Ax : x \in \mathbb{R}^n\}$$

$\exists x$ the whole question is whether one input reaches b

$$Ax = b \text{ is solvable} \iff b \in \mathcal{C}(A)$$

nothing new the right side is the definition of $\mathcal{C}(A)$, read backwards



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.16 range, p. 61. Springer, 2024.

Span and column space

$$\text{SOLVE } c_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ y_3 \end{bmatrix}$$

row 1 $c_1 + 0 = 3$, so $c_1 = 3$

row 2 $0 + c_2 = 2$, so $c_2 = 2$

row 3 $2c_1 + c_2 = 2(3) + 2 = 8$ must equal y_3

TARGET 1

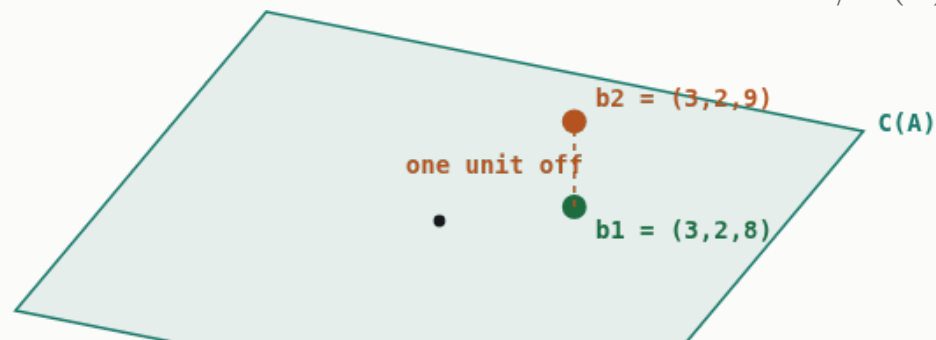
$$y_3 = 8 : 8 = 8 \checkmark$$

TARGET 2

$$y_3 = 9 : 9 \neq 8 \times$$

so $b_1 = 3a_1 + 2a_2 \in \mathcal{C}(A)$

so no c works: $b_2 \notin \mathcal{C}(A)$



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.16 range, p. 61. Springer, 2024.

Span and column space

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

$$Ax = \sum_j x_j a_j$$

x_j the amount taken of column j



$$\mathcal{C}(A) = \text{span}(a_1, \dots, a_n)$$

span every mixture, so every reachable output



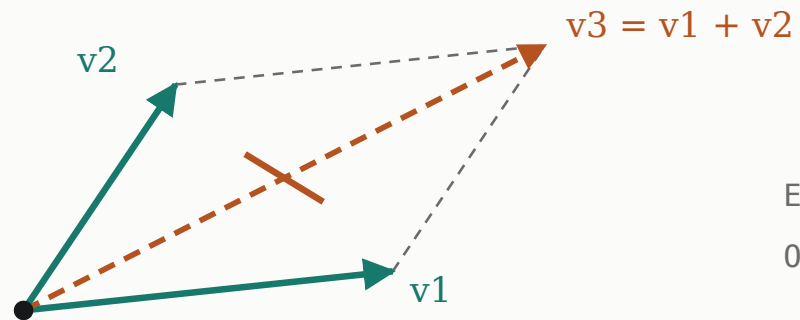
$$Ax = b \text{ solvable} \iff b \in \mathcal{C}(A)$$

our A $a_3 = a_1 + a_2$, so the third column adds nothing to the span

When do different columns carry independent information?

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.15 linearly independent, p. 32. Springer, 2024.

Null Space and Linear Independence



ENOUGH TO REACH?
OR ONE TOO MANY?

Null Space and Linear Independence

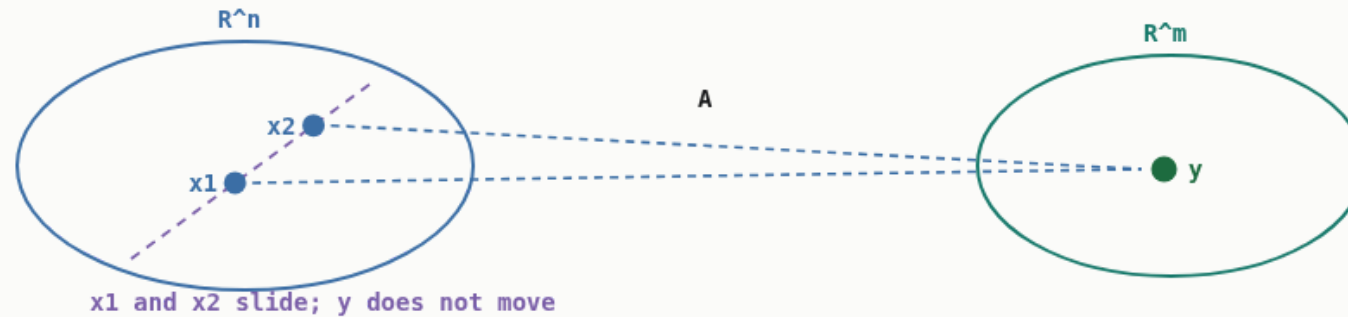
SO FAR $\mathcal{C}(A) =$ all reachable outputs

settled which targets can be hit, and by which mixtures

NOW ASK $Ax_1 = Ax_2, \quad x_1 \neq x_2$

one output but more than one input that reaches it

How can two different inputs produce the same output?



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.14 injective, p. 60; 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

$$Ax_1 = Ax_2$$

$$Ax_1 - Ax_2 = 0$$

$$A(x_1 - x_2) = 0$$

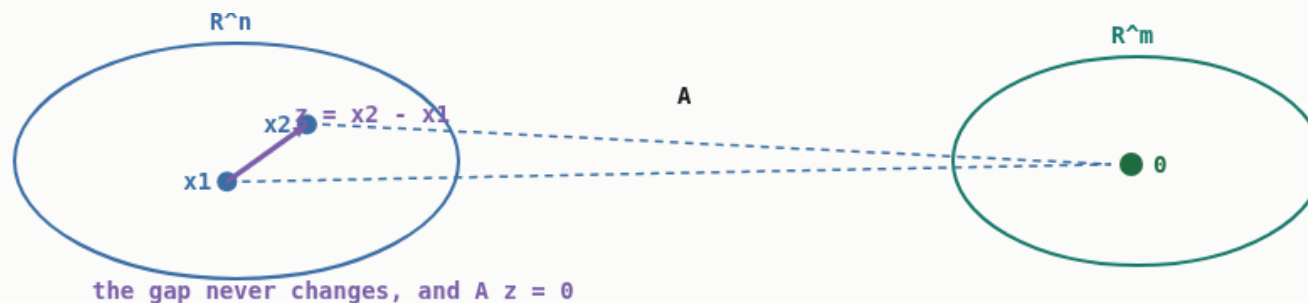
last step linearity, and nothing else, is what pulls A outside

THE DIFFERENCE

$$z = x_1 - x_2 \implies Az = 0$$

z nonzero, since $x_1 \neq x_2$, and A sends it to zero

Every collision $Ax_1 = Ax_2$ is a solution $z = x_1 - x_2$ of $Ax = 0$.



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.1 linear map, p. 52; 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

HOMOGENEOUS SYSTEM

$$Ax = 0$$

homogeneous the right-hand side is the zero vector

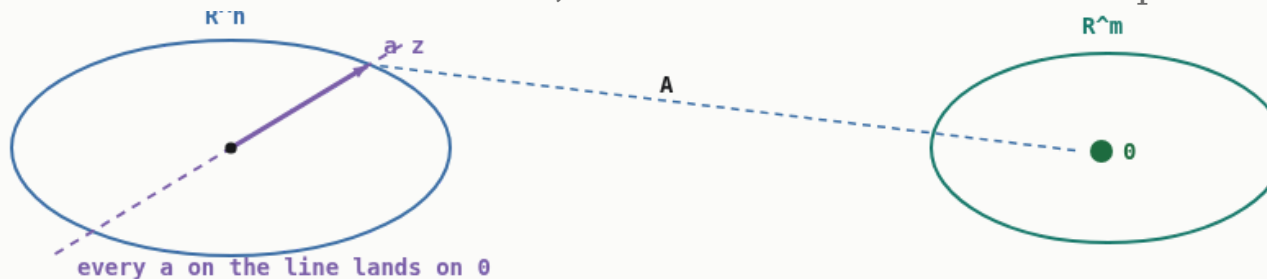
$x = 0$ always a solution, and the uninteresting one

Are there nonzero solutions?

ONE GIVES A LINE

$$Az = 0, z \neq 0 \implies A(\alpha z) = 0 \quad \forall \alpha \in \mathbb{R}$$

αz one invisible direction, so a whole line of invisible inputs



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.26 homogeneous system of linear equations, p. 65. Springer, 2024.

Null Space and Linear Independence

NULL SPACE

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}$$

also called the kernel of A , written $\ker(A)$

$Ax = 0$ the inputs A cannot see

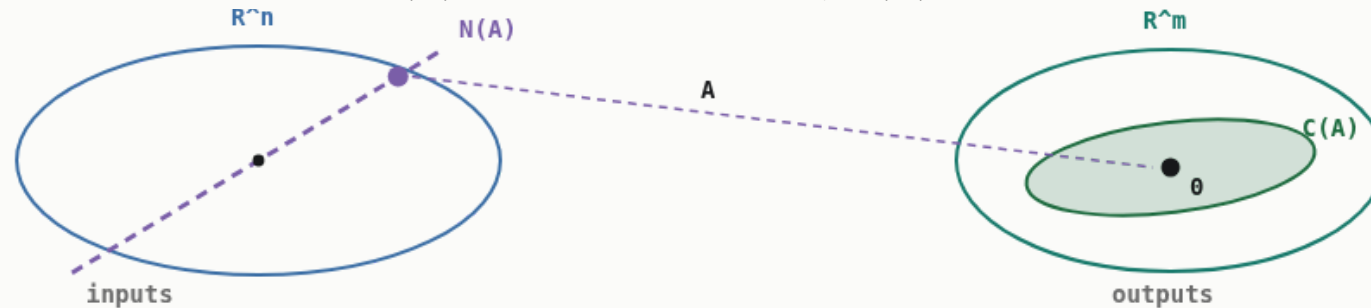
SHAPE

$$\mathcal{N}(A) \subseteq \mathbb{R}^n$$

against

$$\mathcal{C}(A) \subseteq \mathbb{R}^m$$

input side $\mathcal{N}(A)$ lives where x lives, $\mathcal{C}(A)$ where Ax lives



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.11 null space and 3.13 the null space is a subspace, p. 59. Springer, 2024.

Null Space and Linear Independence

$$u, v \in \mathcal{N}(A) \implies Au = 0, Av = 0$$

and $A0 = 0$, so the origin is in $\mathcal{N}(A)$ to begin with

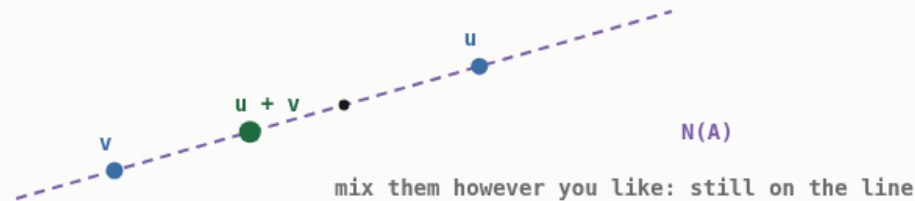
$$A(\alpha u + \beta v) = \alpha Au + \beta Av = 0$$

so $\alpha u + \beta v \in \mathcal{N}(A)$ for every α, β

SUBSPACE

$\mathcal{N}(A)$ is a subspace of $\mathbb{R}^n$

flat in $\mathbb{R}^3$: $\{0\}$, a line, a plane, or all of $\mathbb{R}^3$



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.13 the null space is a subspace, p. 59. Springer, 2024.

Null Space and Linear Independence

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

OUR MATRIX $a_3 = a_1 + a_2 \implies a_1 + a_2 - a_3 = 0$

read it a mixture of the columns, weights 1, 1, -1, giving zero

$$A \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = 0 \implies \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \in \mathcal{N}(A)$$

and the next page solves for all of it

THE WHOLE SET

$$\mathcal{N}(A) = \text{span} \{ (1, 1, -1)^T \}$$

a line through the origin



$a_1 + a_2 - a_3$ walks back to 0
so $(1, 1, -1)$ is in $\mathcal{N}(A)$

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

$$Ax = 0 \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

row 1 $1x_1 + 0x_2 + 1x_3 = 0$, so $x_1 + x_3 = 0$

row 2 $0x_1 + 1x_2 + 1x_3 = 0$

row 3 $2x_1 + x_2 + 3x_3 = 0$ is twice row 1 plus row 2

SOLVE $x_3 = t : x_1 = -t, x_2 = -t, x = t(-1, -1, 1)^T$

why one free unknown

NULL SPACE

$$\mathcal{N}(A) = \text{span}\{(-1, -1, 1)^T\}$$

check $-1 + 1, -1 + 1, -2 - 1 + 3$: all zero

Null Space and Linear Independence

$$z \in \mathcal{N}(A) \implies Az = 0 \implies A(x + z) = Ax + Az = Ax$$

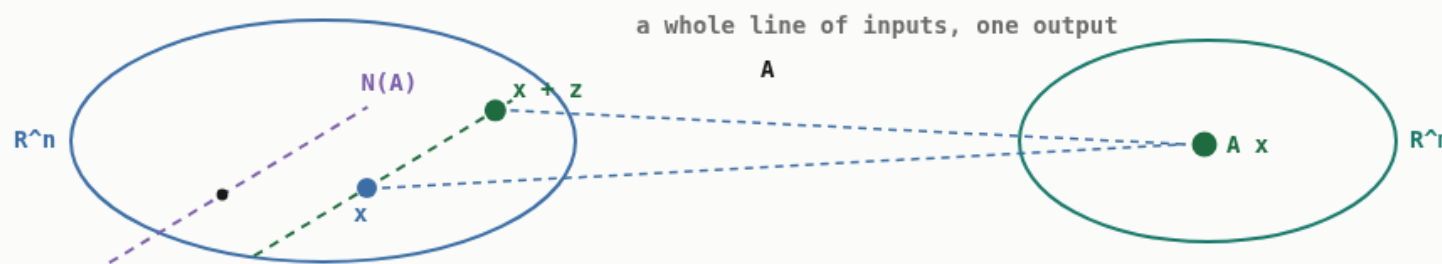
x any input at all, not a special one

WHAT IT MEANS

$$A(x + z) = Ax, \quad z \in \mathcal{N}(A)$$

$x + z$ for our matrix, a line of inputs sharing one output

Null-space directions are invisible to the map.



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.1 linear map, p. 52; 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

ON A ROBOT $\dot{x} = J(q) \dot{q}$

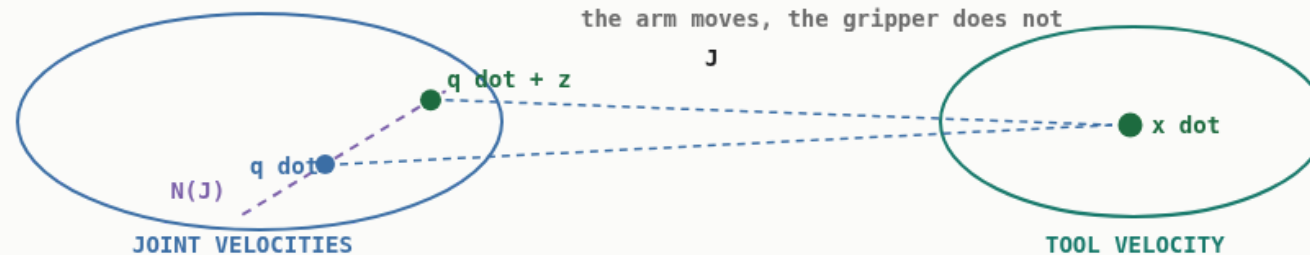
$\dot{q}$ joint velocities, what you command

$\dot{x}$ end-effector velocity, what the world sees

REDUNDANCY

$$z \in \mathcal{N}(J(q)) \implies J(q)(\dot{q} + z) = J(q)\dot{q}$$

z a joint-velocity direction that produces no end-effector velocity



Lynch, K. M. and Park, F. C. *Modern Robotics*, §6.3 inverse velocity kinematics, p. 233. Cambridge University Press, 2017 preprint.

Null Space and Linear Independence

LINEAR INDEPENDENCE

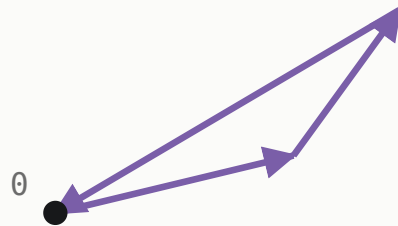
$$c_1 v_1 + \cdots + c_k v_k = 0 \implies c_1 = \cdots = c_k = 0$$

read it the all-zero weights are the only ones that cancel

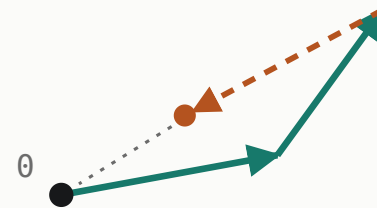
SAME THING

$$\sum_{i=1}^k c_i v_i = 0 \implies c = 0$$

c the whole list of weights, read as one vector



DEPENDENT · THE WALK CLOSES



stops short

INDEPENDENT · IT NEVER CLOSES

Left, a nonzero walk comes home. Right, none does.

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.15 linearly independent, p. 32. Springer, 2024.

Null Space and Linear Independence

LINEAR DEPENDENCE

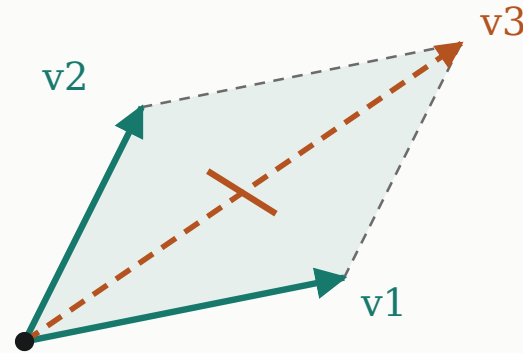
$$\exists c \neq 0 : c_1 v_1 + \cdots + c_k v_k = 0$$

$c \neq 0$ at least one weight is nonzero; say it is c_j

SOLVE FOR THAT ONE

$$v_j = -\frac{1}{c_j} \sum_{i \neq j} c_i v_i$$

so v_j is built from the others, and deleting it leaves the span unchanged



DELETE v_3
SAME PLANE

Strike the diagonal and the plane is untouched.

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.17 linearly dependent and 2.19 linear dependence lemma, p. 33. Springer, 2024.

Null Space and Linear Independence

STACK THEM

$$V = [v_1 \ \cdots \ v_k], \quad Vc = \sum_i c_i v_i$$

Vc the mixture, written as a matrix times its weights

THE ANCHOR

$$\begin{aligned} v_1, \dots, v_k \text{ independent} &\iff \mathcal{N}(V) = \{0\} \\ v_1, \dots, v_k \text{ dependent} &\iff \mathcal{N}(V) \neq \{0\} \end{aligned}$$

one question redundancy in a list is a null space

v1	v2	v3
----	----	----

×

c1
c2
c3

=
●
0

c lives in $\mathcal{N}(V)$

INDEPENDENT MEANS
 $c = 0$ IS THE ONLY ONE

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.15 linearly independent, p. 32; 3.15 injectivity, p. 60. Springer, 2024.

Null Space and Linear Independence

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

OUR MATRIX $a_3 = a_1 + a_2 \implies A(1, 1, -1)^T = 0$

one equation the weights 1, 1, -1 cancel the three columns
NULL SPACE LINEAR DEPENDENCE

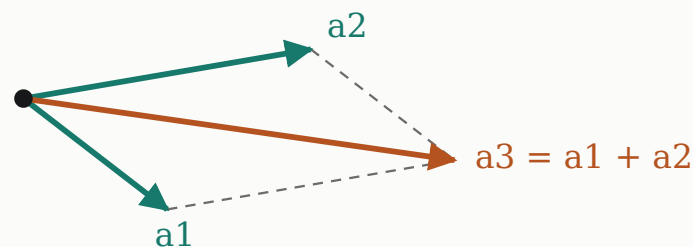
$$(1, 1, -1)^T \in \mathcal{N}(A)$$

a_1, a_2, a_3 dependent

reading an input A cannot see

reading a column you may delete

Same equation, two interpretations.



THE THIRD COLUMN
BUYS NOTHING

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.17 linearly dependent, p. 33; 3.11 null space, p. 59. Springer, 2024.

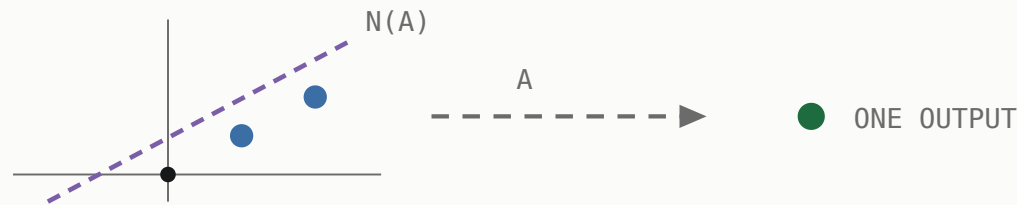
Null Space and Linear Independence

$$\begin{aligned}\mathcal{N}(A) &= \{x : Ax = 0\} \\ z \in \mathcal{N}(A) &\implies A(x + z) = Ax \\ \mathcal{N}(A) \neq \{0\} &\implies \text{input information is lost}\end{aligned}$$

the chain definition, meaning, cost

FOR THE COLUMNS

$$\mathcal{N}(A) = \{0\} \iff \text{columns independent}$$

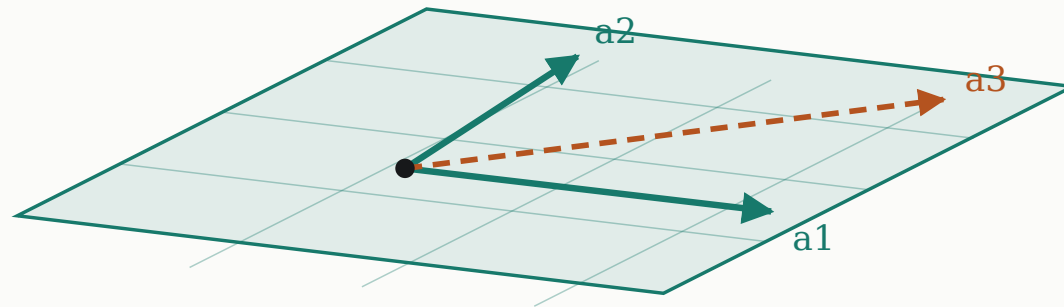


settled whether any direction is redundant

How many independent directions remain?

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank



HOW MANY
SURVIVE?

Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

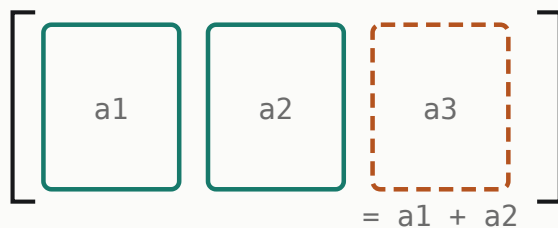
WE KNOW $\mathcal{C}(A)$ what A can produce, $\mathcal{N}(A)$ what A cannot see

and how to tell whether a direction is redundant

NOW ASK

How many independent directions are needed?

our A $a_3 = a_1 + a_2$: three columns, but how many directions?



THREE COLUMNS

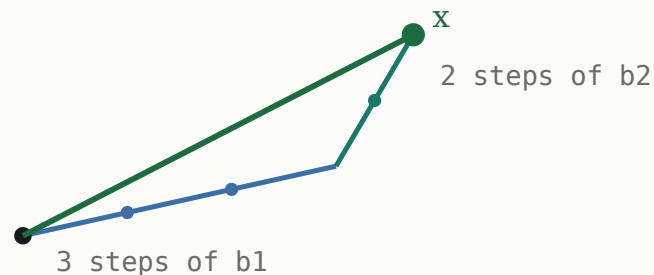
HOW MANY DIRECTIONS?

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank

BASIS

Basis = Spanning + Linear Independence



$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} [x]_B$$

spans $\text{span}(b_1, \dots, b_k) = V$

independent $c_1 b_1 + \dots + c_k b_k = 0 \Rightarrow c_1 = \dots = c_k = 0$

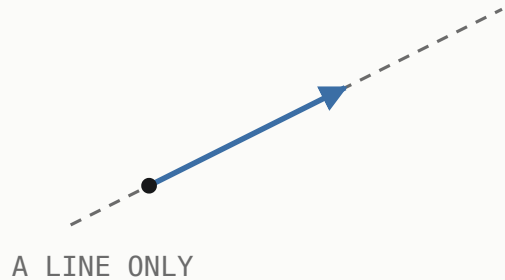
so enough to build everything, with nothing spare

both halves either one alone is not a basis

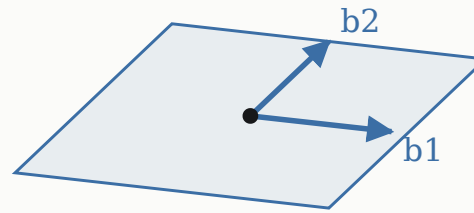
Axler, S. *Linear Algebra Done Right*, 4th ed., 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank

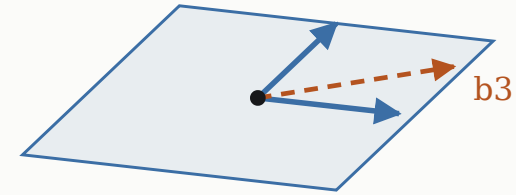
TOO FEW



JUST ENOUGH



TOO MANY



Too few cannot reach; too many give several addresses.

BASIS

enough, but not too many

too few here independent, but does not reach everything

too many reaches everything, but carries a repeat

WORKED EXAMPLE

In $\mathbb{R}^2$: $\{(1,0)\}$ is too few, $\{(1,0), (1,1)\}$ is a basis, adding $(2,1)$ is too many.

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank

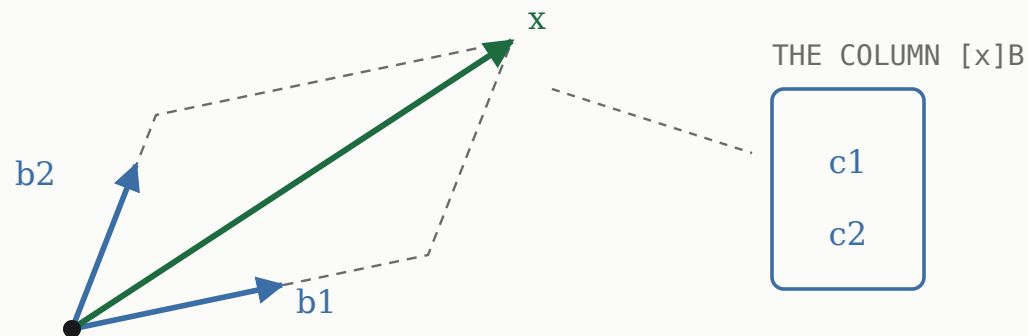
UNIQUE WEIGHTS $x = c_1 b_1 + \cdots + c_k b_k$ in exactly one way

why independence: two different weightings would differ by a nonzero mix that vanishes

COORDINATES

$$[x]_B = (c_1, \dots, c_k)^T, \quad x = B [x]_B$$

one address the vector does not change; only its coordinates do



Axler, S. *Linear Algebra Done Right*, 4th ed., 2.28 criterion for basis, p. 39; 3.73 matrix of a vector, p. 88. Springer, 2024.

Basis, Dimension, and Rank

DIMENSION

$\dim V =$ the number of vectors in any basis of V

a line

a plane

all of $\mathbb{R}^3$

one more independent direction each time

any every basis of the same space has the same length

READING dimension counts independent degrees of freedom

finite every space in this course has a finite basis

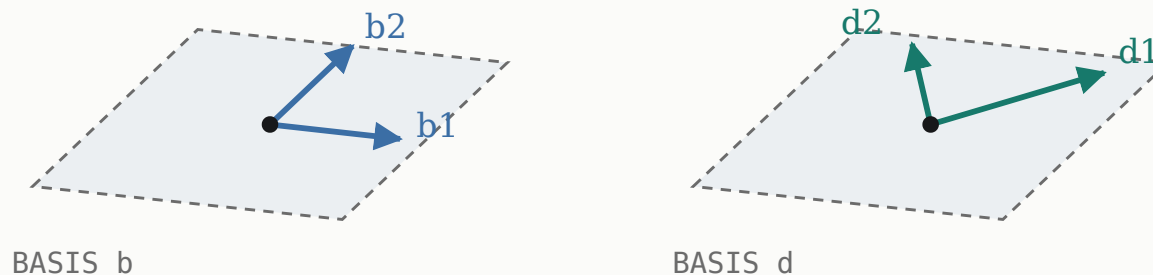
$$\dim\{0\} = 0$$

zero space the empty list is a basis of it

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.35 dimension, p. 44. Springer, 2024.

Basis, Dimension, and Rank

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$



One plane, two bases, both of size two.

SAME SIZE ALWAYS

different bases, different arrows, the same number of them

this plane every basis of it has exactly two vectors

so dimension belongs to the space, not to a choice

Axler, S. *Linear Algebra Done Right*, 4th ed., 2.35 dimension, p. 44. Springer, 2024.

Basis, Dimension, and Rank

$$\text{REDUCE} \quad A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow{r_3 \leftarrow r_3 - 2r_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{r_3 \leftarrow r_3 - r_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

step 1 $(2, 1, 3) - 2(1, 0, 1) = (0, 1, 1)$

step 2 $(0, 1, 1) - (0, 1, 1) = (0, 0, 0)$

pivots columns 1 and 2

$$\text{WHY R} \quad R = EA, \ E \text{ invertible} : Ax = 0 \iff Rx = 0$$

$$\text{so } \mathcal{N}(A) = \mathcal{N}(R)$$

Strang, G. **18.06**, Lecture 7: pivot variables and special solutions. MIT OCW, 2010.

Basis, Dimension, and Rank

PIVOT

the leading nonzero entry of a nonzero row of R

in R pivot positions are read off the echelon form, not off A

pivot variable a variable whose column carries a pivot

free variable a variable whose column does not

$$R = \begin{bmatrix} \textcircled{1} & 0 & \boxed{1} \\ 0 & \textcircled{1} & \boxed{1} \\ 0 & 0 & \boxed{0} \end{bmatrix}$$

x1 x2 x3

pivots

free

HERE x_1, x_2 pivot; x_3 free

count 3 variables, 2 pivots, 1 free

Strang, G. **18.06**, Lecture 7: pivot variables and special solutions. MIT OCW, 2010.

Basis, Dimension, and Rank

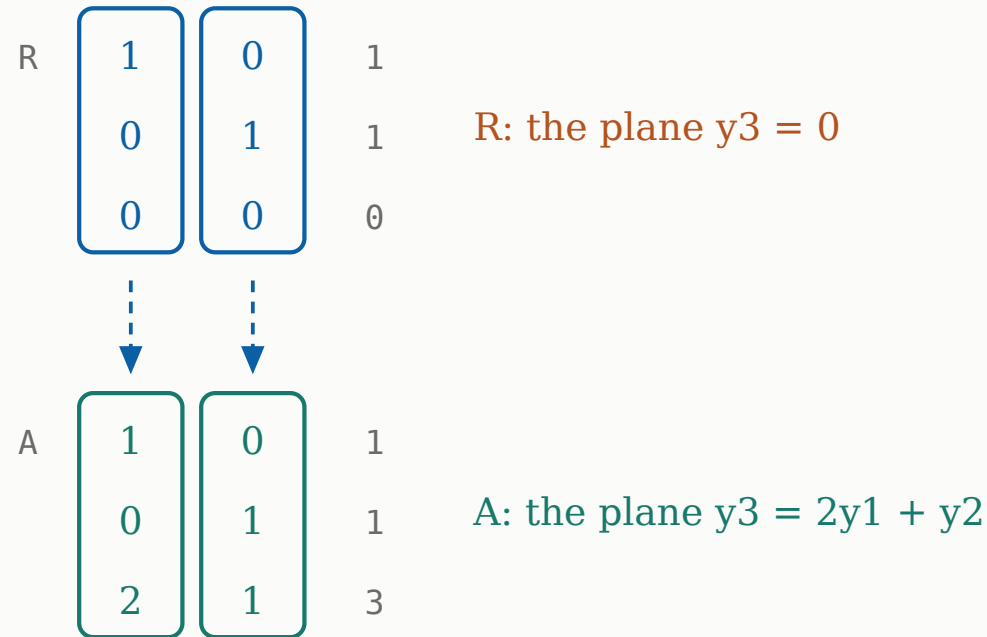
PIVOT COLUMNS

the pivot columns of A are a basis for $\mathcal{C}(A)$

here $\mathcal{N}(A) = \mathcal{N}(R)$, while $\mathcal{C}(A) \neq \mathcal{C}(R)$

R 's columns $(1, 0, 0)^\top, (0, 1, 0)^\top$ span $y_3 = 0$; $a_1 = (1, 0, 2)^\top$ has $y_3 = 2$

A 's columns a_1, a_2 independent, and $a_3 = a_1 + a_2$, so they span $\mathcal{C}(A)$



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.52 column rank, p. 77. Springer, 2024.

Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

ROWS OF R $x_1 + x_3 = 0, \quad x_2 + x_3 = 0, \quad 0 = 0$

legal $\mathcal{N}(A) = \mathcal{N}(R)$

free column 3 has no pivot: $x_3 = t$

SOLVE $x_1 = -t, \quad x_2 = -t \implies x = (-t, -t, t)^T = t(-1, -1, 1)^T$

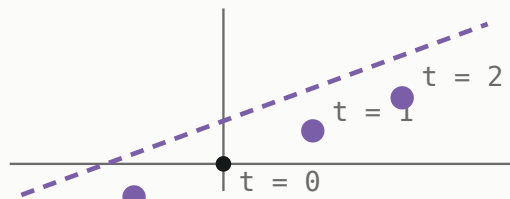
$t=1$ on A: $-1 + 1, -1 + 1, -2 - 1 + 3$, all zero

Basis, Dimension, and Rank

ALL SOLUTIONS $\mathcal{N}(A) = \{ t(-1, -1, 1)^T : t \in \mathbb{R} \}$

why every solution is t times one vector, nothing else

so all multiples of one vector: a span



BASIS AND DIMENSION

$$\mathcal{N}(A) = \text{span}\{(-1, -1, 1)^T\}, \quad \dim \mathcal{N}(A) = 1$$

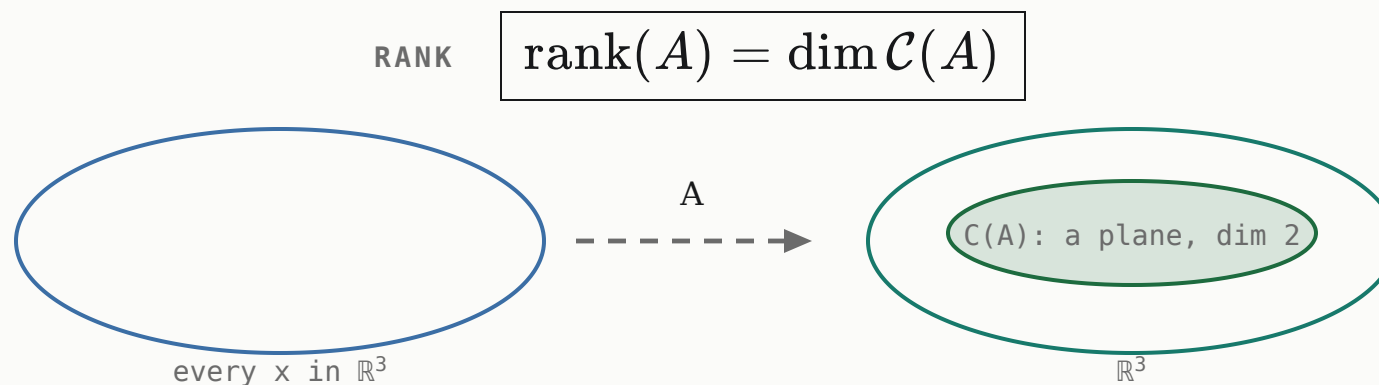
basis one nonzero vector is independent on its own, so it is a basis

count one free variable, one basis vector: $\dim \mathcal{N}(A) = \text{free variables}$

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.11 null space, p. 59; 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$



our A $\dim \mathcal{C}(A) = 2$, so $\text{rank}(A) = 2$

TO COMPUTE

$\text{rank}(A) = \text{the number of pivots}$

reading how many independent output directions A can produce

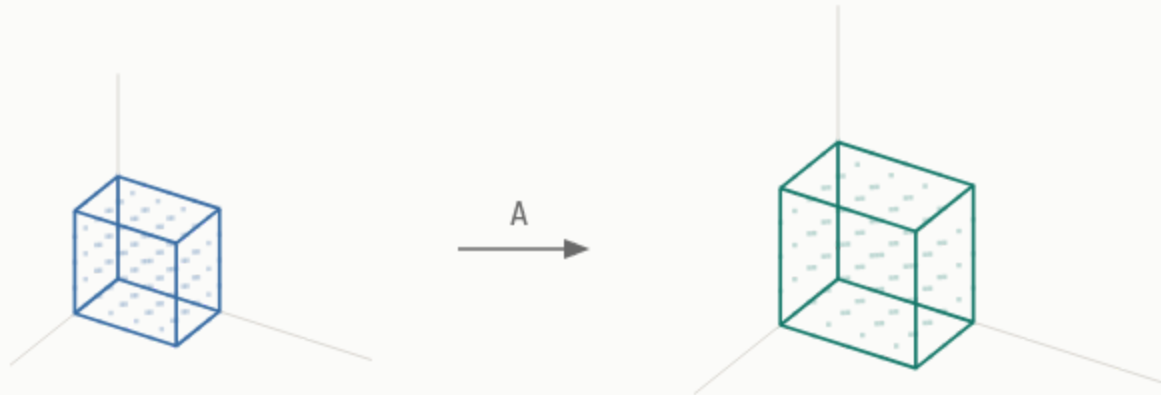
Axler, S. *Linear Algebra Done Right*, 4th ed., 3.58 rank, p. 79; 3.52 column rank, p. 77. Springer, 2024.

Basis, Dimension, and Rank

INTO THREE-SPACE $r = 3$ volume; $r = 2$ plane; $r = 1$ line; $r = 0$ the point $\{0\}$

rank the dimension of the reachable output space

rank 3 · the image is a volume



every input, a solid cube

The input cube never changes. Each direction the matrix loses costs the image a whole dimension.

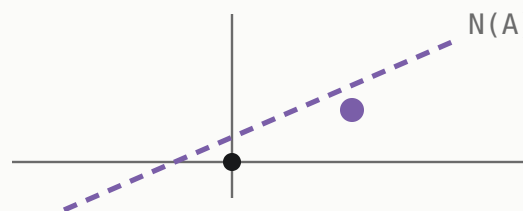
Sanderson, G. *Essence of Linear Algebra*, ch. 7. 3Blue1Brown, 2016.

Basis, Dimension, and Rank

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

NULLITY

$$\text{nullity}(A) = \dim \mathcal{N}(A)$$



ONE FREE DIRECTION

$$\text{nullity}(A) = 1$$

our A $\mathcal{N}(A) = \text{span}\{(-1, -1, 1)^T\}$, so $\text{nullity}(A) = 1$

READING independent input directions lost by A

the pair rank counts what survives, nullity counts what disappears

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Basis, Dimension, and Rank

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

RANK-NULLITY

$$\text{rank}(A) + \text{nullity}(A) = n$$

n the number of columns, the input dimension

reach $\dim \mathcal{C}(A) = r$: directions that survive

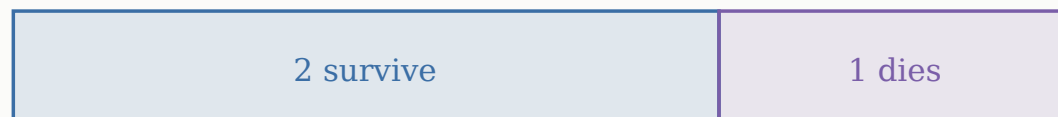
loss $\dim \mathcal{N}(A) = n - r$: directions that disappear

our A $2 + 1 = 3$

a choice $n - r$ basis vectors in $\mathcal{N}(A)$, r whose images span $\mathcal{C}(A)$

$$r = 2$$

$$n - r = 1$$



$$n = 3 \text{ columns of } A$$

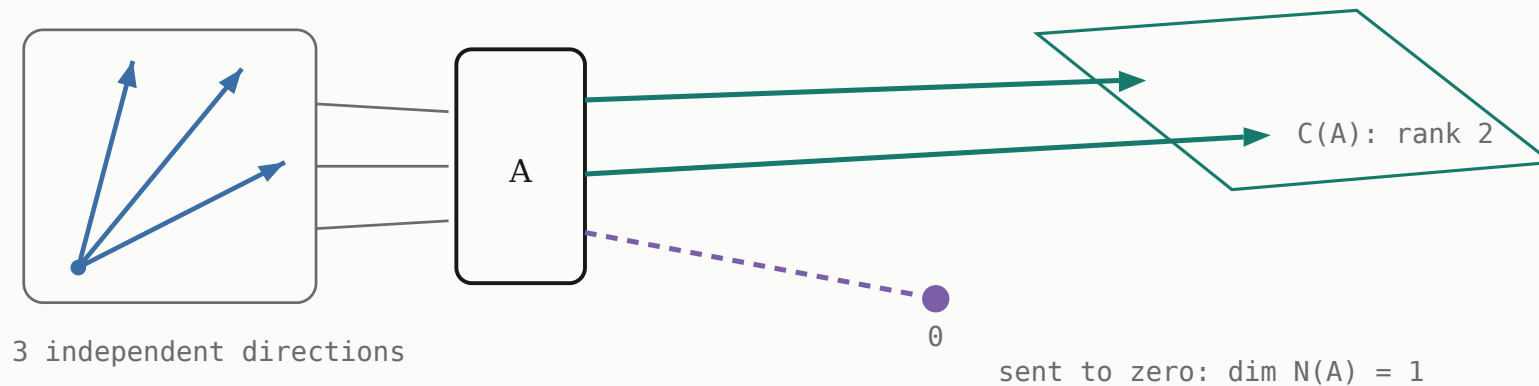
Axler, S. *Linear Algebra Done Right*, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Basis, Dimension, and Rank

THE MAP $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $n = r + (n - r)$

r directions that survive

$n - r$ directions that vanish



Two survivors, one casualty, three directions accounted for.

so **rank-nullity counts the input side**

not m the null space is not in the output space

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Basis, Dimension, and Rank

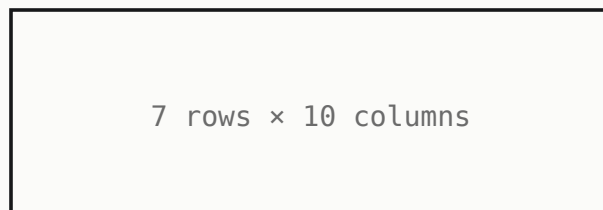
EXAM QUESTION $B \in \mathbb{R}^{7 \times 10}$, $\text{rank}(B) = 6$

ask $\dim \mathcal{C}(B)$, $\dim \mathcal{N}(B)$, and the number of free variables

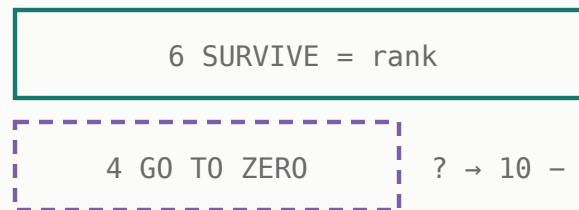
$$\dim \mathcal{C}(B) = 6, \quad \dim \mathcal{N}(B) = 10 - 6 = 4, \quad 4 \text{ free variables}$$

the trap use the 10, not the 7

B



THE TEN INPUT DIRECTIONS



USE THE 10,
NOT THE 7

$$? \rightarrow 10 - 6$$

$$\dim \mathcal{N}(B) = n - r = 10 - 6 = 4.$$

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Basis, Dimension, and Rank

FULL COLUMN RANK

$$r = n$$

then $\mathcal{N}(A) = \{0\}$: no input direction is lost; needs $n \leq m$

all n inputs survive

$$\mathcal{N}(A) = \{0\}$$

FULL ROW RANK

$$r = m$$

then $\mathcal{C}(A) = \mathbb{R}^m$: every output is reachable; needs $m \leq n$

all m outputs reached

$$\mathcal{C}(A) = \mathbb{R}^m$$

One is about losing nothing. The other is about reaching everything.

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Basis, Dimension, and Rank

WHERE WE ARE $\dim \mathcal{C}(A) = r, \quad \dim \mathcal{N}(A) = n - r$



so the three questions become countable

is b reachable? $b \in \mathcal{C}(A)$

is x unique? $\mathcal{N}(A) = \{0\}$

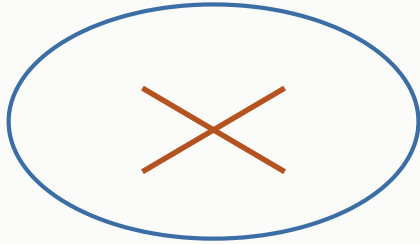
how many free directions? $n - r$

row two uniqueness is a question only once a solution exists

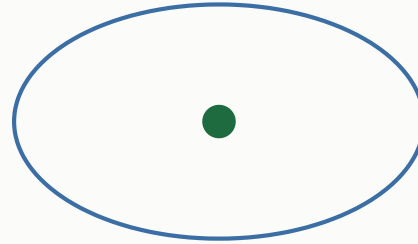
Axler, S. *Linear Algebra Done Right*, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Linear Systems and Solution Sets

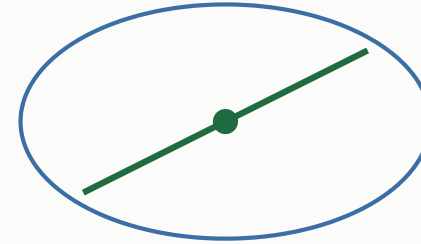
NONE



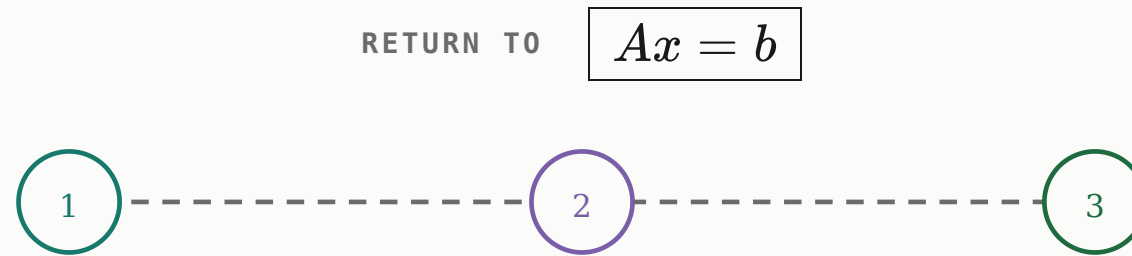
EXACTLY ONE



INFINITELY MANY



Linear Systems and Solution Sets



1 **existence** does a solution exist?

2 **uniqueness** if it exists, is it the only one?

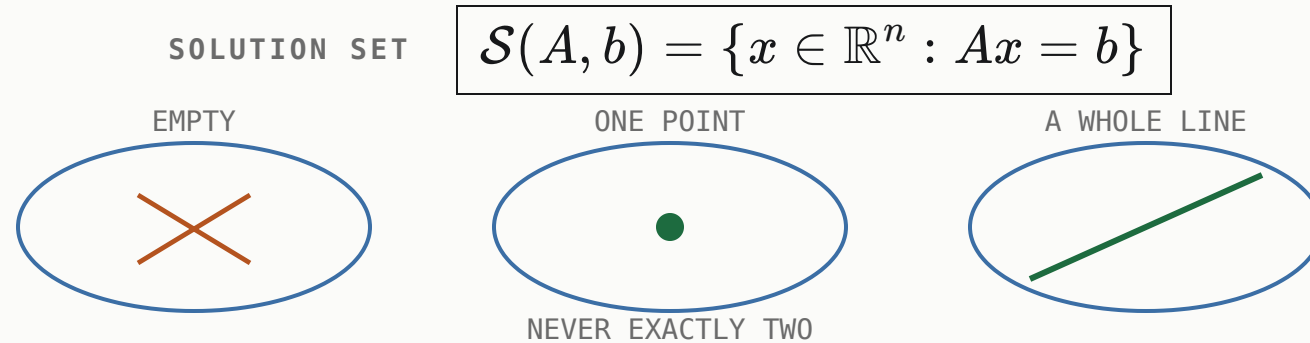
3 **structure** what are all of them?

WITH $\mathcal{C}(A)$ and $\mathcal{N}(A)$

nothing new both spaces are already built and measured

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.16 range, p. 61; 3.11 null space, p. 59. Springer, 2024.

Linear Systems and Solution Sets



a **solution** any x that produces the requested b

consistent $\mathcal{S}(A, b) \neq \emptyset$; otherwise inconsistent

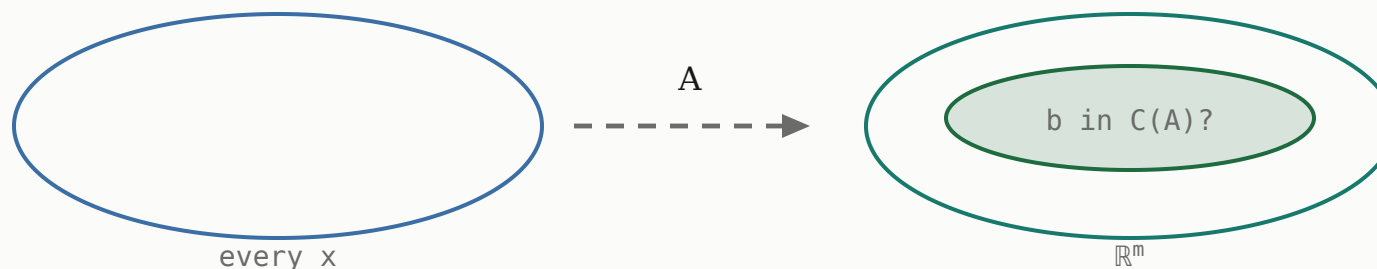
THREE POSSIBILITIES empty, one point, infinitely many

never two two solutions force infinitely many: a line, a plane, or higher

Linear Systems and Solution Sets

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

RECALL $\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\}$



so b is reachable exactly when some x reaches it

EXISTENCE $Ax = b \text{ is consistent} \iff b \in \mathcal{C}(A)$

one sentence the column space controls existence

WORKED EXAMPLE

$\mathcal{C}(A)$ is $y_3 = 2y_1 + y_2$: $(3, 2, 8)$ passes since $2(3) + 2 = 8$; $(3, 2, 9)$ fails.

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.16 range, p. 61. Springer, 2024.

Linear Systems and Solution Sets

SUPPOSE $Ax_1 = b, \quad Ax_2 = b \implies A(x_1 - x_2) = 0 \implies x_1 - x_2 \in \mathcal{N}(A)$

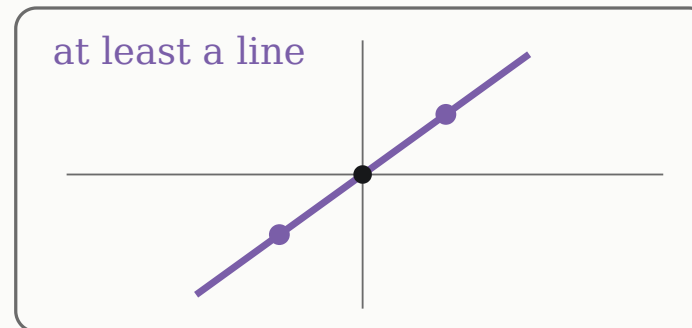
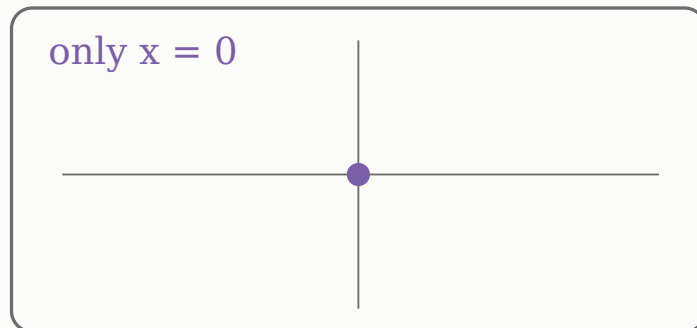
so if $\mathcal{N}(A) = \{0\}$ then $x_1 = x_2$

UNIQUENESS

$\mathcal{N}(A) = \{0\} \implies$ at most one solution

careful at most one, not one

if consistent then it means exactly one



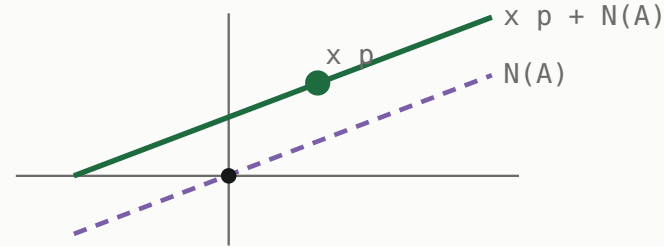
Either only the origin, or at least a whole line of erased inputs.

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.15 injectivity, p. 60. Springer, 2024.

Linear Systems and Solution Sets

PARTICULAR

$$Ax_p = b \text{ for one particular } x_p$$



ONE SOLUTION, THEN
EVERYTHING A IGNORES

any one it does not matter which solution you found

GENERAL

$$\mathcal{S}(A, b) = x_p + \mathcal{N}(A) = \{x_p + z : z \in \mathcal{N}(A)\}$$

recipe find one solution, then add everything A ignores

Linear Systems and Solution Sets

EACH ONE WORKS

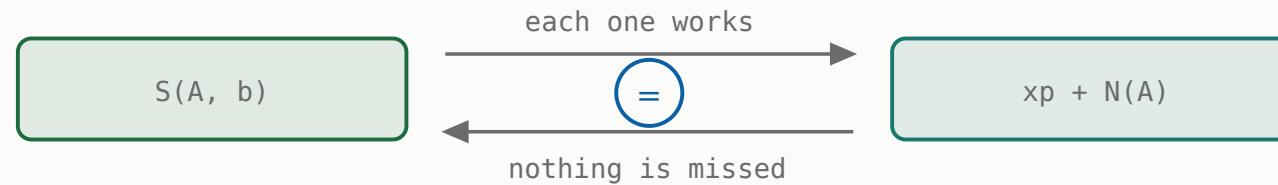
$$A(x_p + z) = b + 0 = b$$

uses linearity and $Az = 0$

NOTHING IS MISSED

$$Ax = b \Rightarrow A(x - x_p) = 0$$

so $x = x_p + z$ with $z = x - x_p \in \mathcal{N}(A)$



EQUALITY

$$\mathcal{S}(A, b) = x_p + \mathcal{N}(A)$$

both ways one inclusion each, which is what an equality of sets needs

Linear Systems and Solution Sets

$$\text{AUGMENT} \quad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 3 & 8 \end{array} \right] \xrightarrow{r_3 \leftarrow r_3 - 2r_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{array} \right] \xrightarrow{r_3 \leftarrow r_3 - r_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

step 1 $(2, 1, 3|8) - 2(1, 0, 1|3) = (0, 1, 1|2)$

step 2 $(0, 1, 1|2) - (0, 1, 1|2) = (0, 0, 0|0)$

row 3 $0 = 0$: consistent

Linear Systems and Solution Sets

READ OFF $x_1 + x_3 = 3, \quad x_2 + x_3 = 2, \quad x_3 = t \implies x_1 = 3 - t, \quad x_2 = 2 - t$

free no pivot in column 3: choose x_3

$t = 0$ one particular solution, $x_p = (3, 2, 0)^\top$

ALL OF THEM

$$x = (3, 2, 0)^\top + t(-1, -1, 1)^\top, \quad t \in \mathbb{R}$$

$t = 1$ $(2, 1, 1)^\top$: $2 + 1 = 3, 1 + 1 = 2, 4 + 1 + 3 = 8$

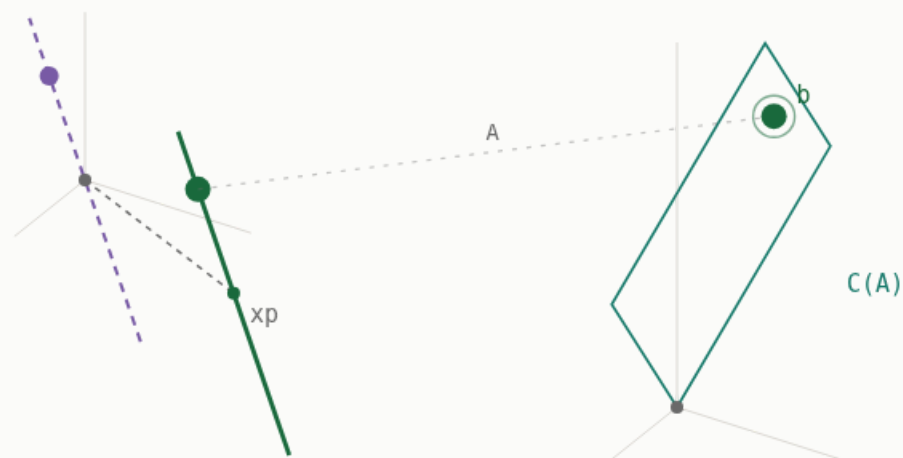
$b = (3, 2, 9)^\top$ same steps: $9 - 6 = 3, 3 - 2 = 1$, so row 3 reads $0 = 1$: no solution

Linear Systems and Solution Sets

AFFINE SET

$$x_0 + S = \{x_0 + s : s \in S\} \text{ for a subspace } S$$

here $\mathcal{S}(A, b) = x_p + \mathcal{N}(A)$: the null space, translated
through 0 ? only if $b = 0$; then $\mathcal{S} = \mathcal{N}(A)$, a subspace



$\mathcal{N}(A)$ through the origin

$x = x_p + t z$, the same line moved

b never moves

The null space through the origin, the same line moved to a particular solution, and an output that never moves.

Axler, S. *Linear Algebra Done Right*, 4th ed., 1.34 conditions for a subspace, p. 18. Springer, 2024.

Linear Systems and Solution Sets

NOTATION $r = \text{rank}(A)$

unique when consistent, means $r = n$

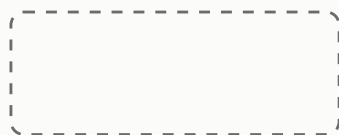
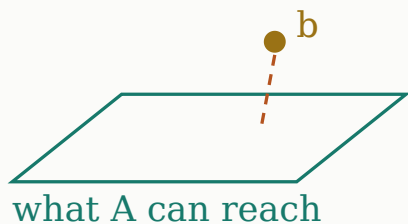
CASES

$b \notin \mathcal{C}(A) : \text{none}$

$b \in \mathcal{C}(A), \mathcal{N}(A) = \{0\} : \text{one}$

$b \in \mathcal{C}(A), \dim \mathcal{N}(A) > 0 : \text{infinitely many}$

free $n - r$ directions



none



exactly one

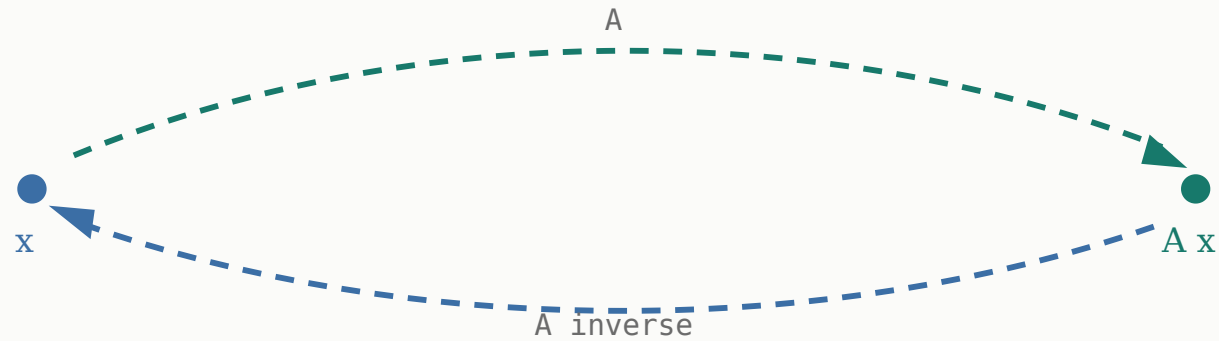


infinitely many

Top: the demand. Bottom: the solutions.

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Invertibility and Singular Matrices



Invertibility and Singular Matrices

DEMAND $\mathcal{C}(A) = \mathbb{R}^m$ and $\mathcal{N}(A) = \{0\}$

reach $r = m$: nothing is missed

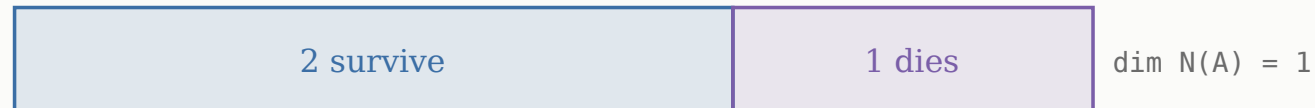
no loss $r = n$ by rank-nullity

so

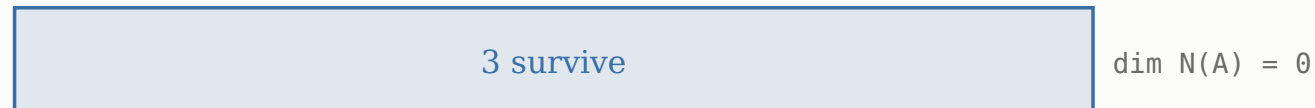
$$m = n \text{ and } r = n$$

square a conclusion, not a hypothesis

$r = 2$



$r = 3$, nothing lost



$n = 3$ both rows

Both cases have three input directions to spend. Only the top spends one on the null space.

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Invertibility and Singular Matrices

INVERSE

$$AA^{-1} = A^{-1}A = I$$

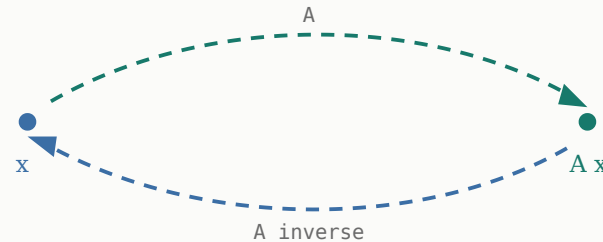
shapes $A, A^{-1} \in \mathbb{R}^{n \times n}$ and $I = I_n$

both sides two-sided by definition; for square A either half implies the other

SINGULAR

A singular $\iff A$ has no inverse

nonsingular the same as invertible



WORKED EXAMPLE

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Axler, S. **Linear Algebra Done Right**, 4th ed., 3.79 identity, p. 90; 3.80 invertible and the word singular, p. 91; 3.60 inverse is unique, p. 82. Springer, 2024.

Invertibility and Singular Matrices

IN GENERAL A invertible $\iff A$ injective and surjective

injective $Ax = Ay \Rightarrow x = y$

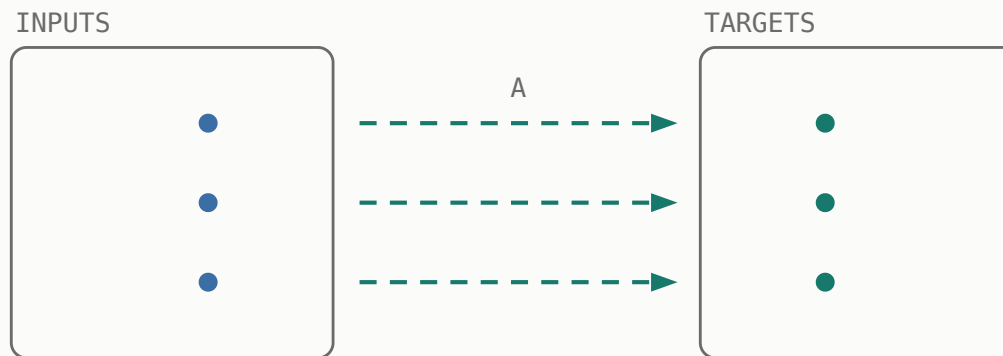
surjective $\mathcal{C}(A) = \mathbb{R}^m$

IF A IS SQUARE

$$\mathcal{N}(A) = \{0\} \iff \mathcal{C}(A) = \mathbb{R}^n \iff A \text{ invertible}$$

saving check one side, the other follows

why rank-nullity: $r = n$ makes both true at once



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.63 invertibility, p. 83; 3.65 injectivity equivalent to surjectivity, p. 84. Springer, 2024.

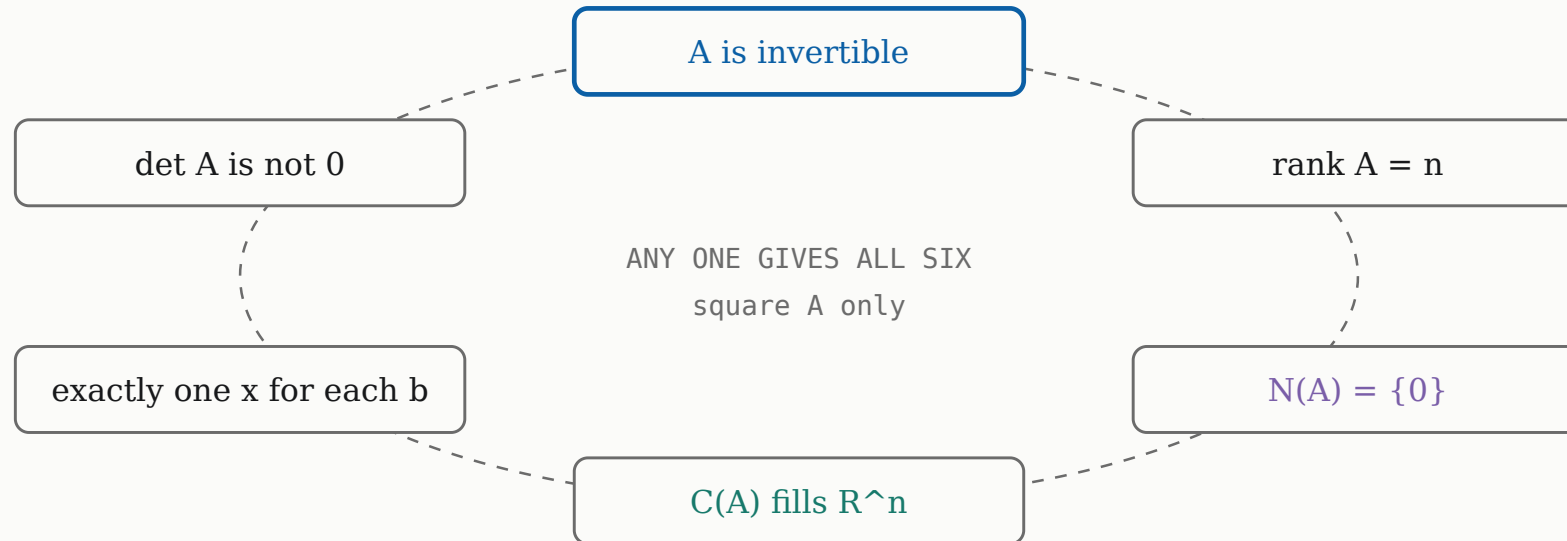
Invertibility and Singular Matrices

SQUARE A

$$\begin{aligned} A \text{ invertible} &\iff r = n \iff \mathcal{N}(A) = \{0\} \\ &\iff \text{columns independent} \iff \mathcal{C}(A) = \mathbb{R}^n \end{aligned}$$

and $\iff Ax = b$ has one solution for every b

det a preview; the determinant is a later lecture



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.63 invertibility, p. 83; 3.65, p. 84; 3.21, p. 62. Springer, 2024.

Invertibility and Singular Matrices

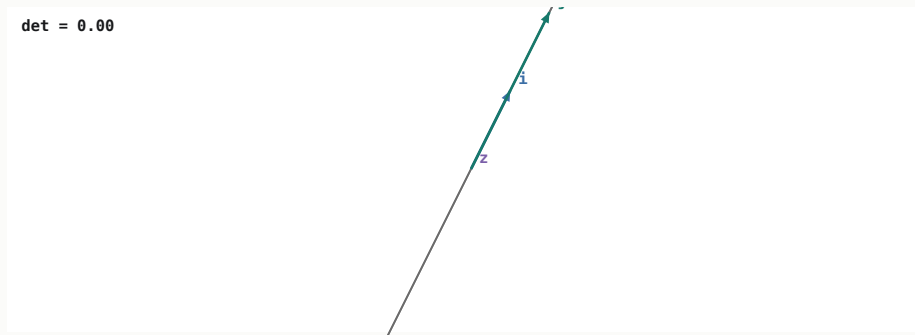
SINGULAR

$$\mathcal{N}(B) \neq \{0\} \iff B \text{ singular}$$

B square and rank deficient

WORKED EXAMPLE

$B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has $r = 1$, and $B(2, -1)^T = 0$: a nonzero input lands on zero.



The unit square flattens; the area, and the inverse, are gone.

Sanderson, G. *Essence of Linear Algebra*, ch. 7. 3Blue1Brown, 2016 · Axler, 4th ed., 3.80, p. 91.

Invertibility and Singular Matrices

PREVIEW · LEAST SQUARES

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}, \quad X^T X = \begin{bmatrix} 1 + 4 + 9 & 2 + 8 + 18 \\ 2 + 8 + 18 & 4 + 16 + 36 \end{bmatrix} = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}$$

the fit

$$\min_{\beta} \|X\beta - y\|^2 \Rightarrow X^T X \beta = X^T y$$

entries $(X^T X)_{ij}$ is column i dotted with column j ; the off-diagonal pair agree

Goodfellow, I., Bengio, Y. and Courville, A. **Deep Learning**, ch. 2. MIT Press, 2016.

Invertibility and Singular Matrices

SINGULAR $X \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 - 2 \\ 4 - 4 \\ 6 - 6 \end{bmatrix} = 0 \implies X^\top X \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 0$

collinear column 2 is twice column 1: $(2, -1)^\top \in \mathcal{N}(X)$

NULL SPACES
$$X^\top X z = 0 \implies \|X z\|^2 = z^\top X^\top X z = 0 \implies X z = 0$$

so $\mathcal{N}(X^\top X) = \mathcal{N}(X)$

the fit $\beta + t(2, -1)^\top$ gives the same $X\beta$: not identifiable

Invertibility and Singular Matrices

AT A SINGULARITY

$J(q)$ loses rank $\implies$ a tool direction is lost

rank drop $\mathcal{C}(J)$ shrinks: some $\dot{x}$ has no $\dot{q}$

nearby joint rates grow without bound for tool directions along the lost one

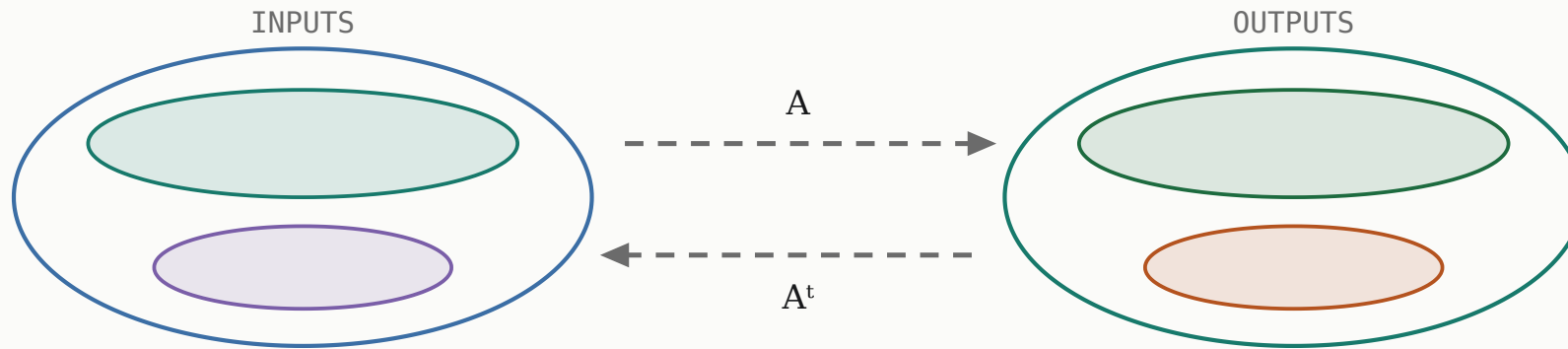
NEXT $\mathcal{C}(A), \mathcal{N}(A), \mathcal{C}(A^T), \mathcal{N}(A^T)$

one matrix four spaces, one rank

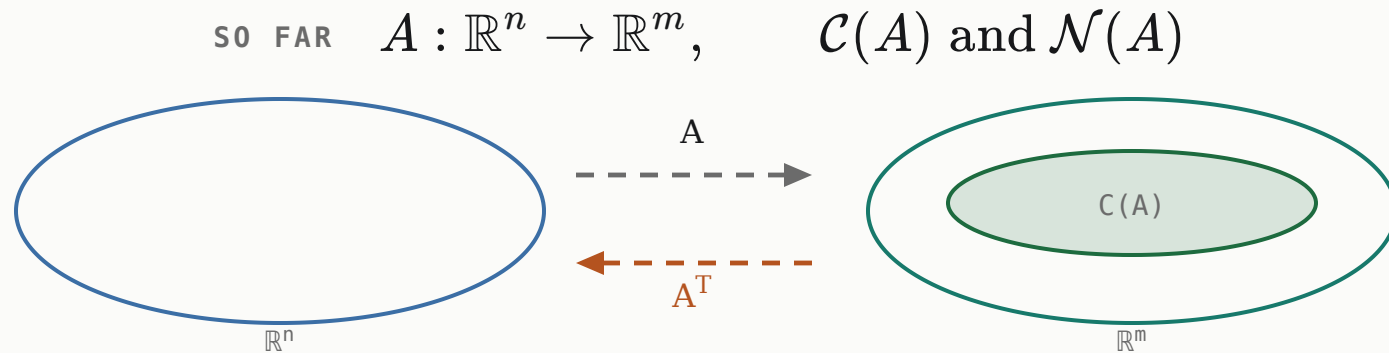


Lynch, K. M. and Park, F. C. **Modern Robotics**, §5.3 singularity analysis, p. 191. Cambridge University Press, 2017 preprint.

The Four Fundamental Subspaces



The Four Fundamental Subspaces



but $A^T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a linear map too

SO ASK

what spaces come from A^T ?

the answer $\mathcal{C}(A^T)$ and $\mathcal{N}(A^T)$: two more, for free

The Four Fundamental Subspaces

NEW NAMES

$$\text{Row}(A) = \mathcal{C}(A^T), \quad \mathcal{N}(A^T) = \{y : A^T y = 0\}$$

$\mathbb{R}^n$ · INPUTS

row space $\mathcal{C}(A^T)$

null space $\mathcal{N}(A)$

$\mathbb{R}^m$ · OUTPUTS

column space $\mathcal{C}(A)$

left null $\mathcal{N}(A^T)$

left null $A^T y = 0$ is $y^T A = 0^T$: y multiplies from the left

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.54 transpose, p. 77; 3.11 null space, p. 59. Springer, 2024.

The Four Fundamental Subspaces

OUTPUT SIDE

$$\mathcal{C}(A) = \text{span}\{(1, 0, 2)^\top, (0, 1, 1)^\top\}$$

$$\mathcal{N}(A^\top) = \text{span}\{(-2, -1, 1)^\top\}$$

INPUT SIDE

$$\text{Row}(A) = \text{span}\{(1, 0, 1)^\top, (0, 1, 1)^\top\}$$

$$\mathcal{N}(A) = \text{span}\{(-1, -1, 1)^\top\}$$

$$\begin{array}{rcl} \text{row 1} & 1 & 0 & 1 \\ \text{row 2} & 0 & 1 & 1 \\ \text{row 3} & 2 & 1 & 3 \end{array} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} 2r_1 + r_2 = r_3 \\ \text{so } -2r_1 - r_2 + r_3 = 0 \end{array}$$

$$\text{so } \boxed{y = (-2, -1, 1)^\top, \quad A^\top y = 0}$$

read it the row weights

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.57 column rank equals row rank, p. 78. Springer, 2024.

The Four Fundamental Subspaces

RANK EQUALITY

$$\text{rank}(A) = \text{rank}(A^T) = r$$

$$r = 2$$

$$m - r = 1$$

the output side: $r = 2$, $m - r = 1$

so column rank equals row rank

ALL FOUR

$$\begin{aligned} \dim \mathcal{C}(A) &= r & \dim \mathcal{N}(A) &= n - r \\ \dim \mathcal{C}(A^T) &= r & \dim \mathcal{N}(A^T) &= m - r \end{aligned}$$

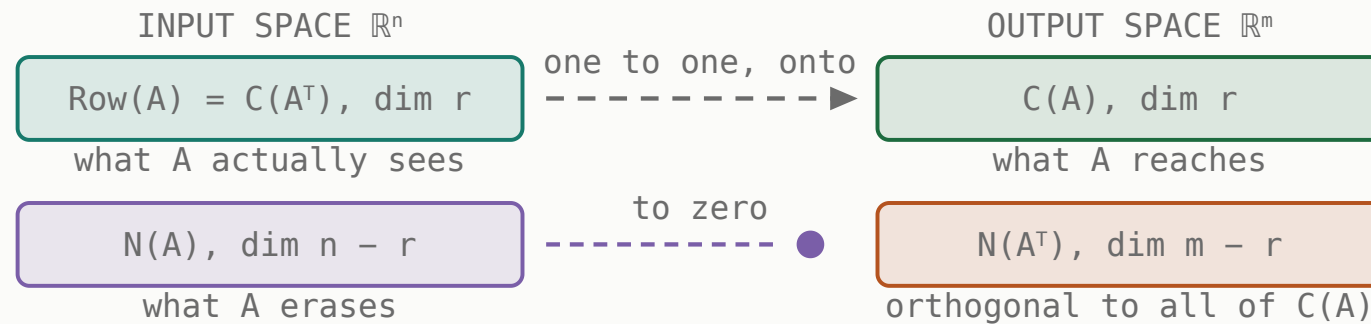
two sums $r + (n - r) = n$ and $r + (m - r) = m$

WORKED EXAMPLE

Our A : $r = 2$, so $2 + 1 = 3$ columns and $2 + 1 = 3$ rows.

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.57 column rank equals row rank, p. 78; 3.21, p. 62. Springer, 2024.

The Four Fundamental Subspaces



LECTURE 1 IN ONE LINE

$$\dim = r, n - r, r, m - r$$

later the two pairs form orthogonal-complement decompositions

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.21, p. 62; 3.57, p. 78. Springer, 2024.

Courses, books and lectures

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3. Axler, S. [*Linear Algebra Done Right*](#), 4th ed., 2024. Open access.
4. Goodfellow, Bengio and Courville. [*Deep Learning*](#), MIT Press, 2016, ch. 2.
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Papers and books

6. Strang, G. The fundamental theorem of linear algebra. *Amer. Math. Monthly*, 1993.
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8. Lipman, Y. et al. Flow Matching for Generative Modeling. *ICLR*, 2023. [arXiv:2210.02747](#).
9. Lynch and Park. *Modern Robotics*, §5.3, §6.3, 2017.
10. Barfoot, T. D. *State Estimation for Robotics*, ch. 4. Cambridge, 2017.